

Theory of Condensed Matter

Course notes (V. Cheianov)

Francisco Gómez Serrano

Contents

1	Introduction and the harmonic oscillator	4
1.1	Harmonic oscillator	4
1.2	Extended Wigner–Heisenberg algebra	6
1.3	Exercises	6
2	Dielectric crystals and phonons	7
2.1	One-dimensional monoatomic crystal	8
2.2	Lattice Fourier transform	8
2.3	Normal modes of the monoatomic crystal	9
2.4	Exercises	11
3	Thermodynamics and linear response of the harmonic crystal	12
3.1	Thermodynamics	12
3.2	Dynamics: linear response and the Kubo formula	14
4	Polyatomic and three-dimensional lattices	16
4.1	One-dimensional diatomic crystal	16
4.2	Three-dimensional lattices	17
4.3	Polyatomic one-dimensional lattice	18
4.4	Polyatomic three-dimensional lattice	20
5	Processes: Fermi’s golden rule and neutron scattering	20
5.1	Fermi’s golden rule	21
5.2	Application: neutron scattering experiment	21
5.3	Inelastic scattering	22
6	Superfluidity	23
6.1	Superfluid helium: experiments	23
6.2	The Landau criterion (1941)	25
6.3	The ideal Bose gas	26

7	Second quantisation	27
7.1	Independent bosons	27
7.2	Second quantisation	28
7.3	Quantum field	29
8	Bogoliubov's theory of superfluidity	29
8.1	Interacting bosons	29
8.2	Semiclassical condensate	30
8.3	Bogoliubov's mean-field Hamiltonian	31
8.4	Condensate depletion	32
9	The Gross–Pitaevskii equation	32
9.1	Semiclassical theory	32
9.2	The equations of motion	33
9.3	Hard and soft modes	34
9.4	Vortices	34
10	Fermions	35
10.1	Independent fermions	35
10.2	Second quantisation	35
10.3	Ground state and elementary excitations	36
10.4	Thermodynamics	36
10.5	Ideal Fermi gas	37
11	Weakly interacting Fermi gas	38
11.1	The Hamiltonian	39
11.2	Hartree–Fock theory	39
11.3	The ground state	40
11.4	Generalisation: the Landau theory	40
11.5	The Wilson ratio	40
12	Superconductivity	41
12.1	The Landau–Ginzburg theory	41
13	The BCS theory	43
13.1	The Cooper phenomenon	43
13.2	The source of attraction	43
13.3	The electron–phonon interaction	44
13.4	The effective interaction strategy	44

14 The BCS theory (continued)	45
14.1 The broken-symmetry state	45
14.2 BCS mean-field theory	46
14.3 Self-consistency condition	47
Revision notes	47

1 Introduction and the harmonic oscillator

Condensed matter is the study of interacting particles. “Large” means more than about ten particles; in condensed matter typically $\gtrsim 10^4$.

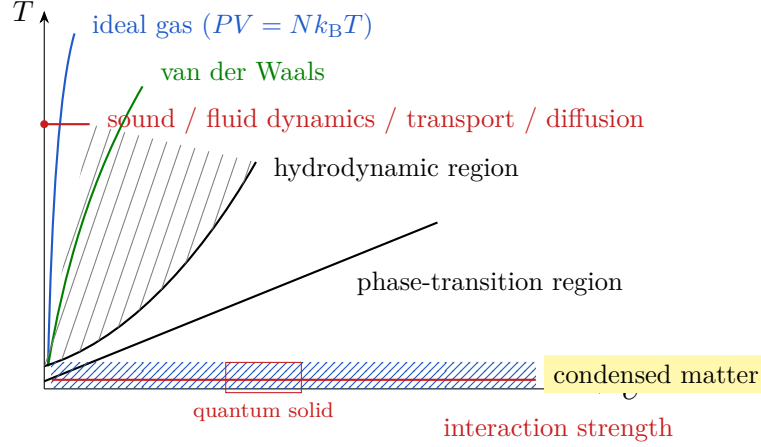


Figure 1.1: Schematic “phase diagram” of matter in the plane (interaction strength U , temperature T). The hydrodynamic and phase-transition regions are not present in non-interacting systems. The van der Waals equation of state is $(P + aN^2/V^2)(V - bN) = Nk_B T$.

As $U \rightarrow 0$ the behaviour is critical (there is a discontinuity): the **mean free path** λ_{mf} diverges (it is $\sim$ the interaction range). Comparing with the system size L :

- $\lambda_{\text{mf}}(U) > L$: ideal system;
- $\lambda_{\text{mf}}(U) < L$: one can **coarse-grain** the system into cells.

Thermal estimate of the de Broglie wavelength $\lambda \sim \hbar/p$:

$$\frac{p^2}{2m} \sim E \sim T, \quad p \sim \frac{\hbar}{\lambda} \quad \Rightarrow \quad \frac{\hbar^2}{m\lambda^2} \sim T.$$

For strong interactions, λ_{mf} is the distance between the particles. Then

- $\lambda_{\text{mf}} \gg \lambda$: classical system;
- $\lambda_{\text{mf}} \lesssim \lambda$: quantum system, i.e.

$$\frac{1}{\lambda_{\text{mf}}} \sqrt{\frac{\hbar^2}{mT}} \gtrsim 1.$$

1.1 Harmonic oscillator

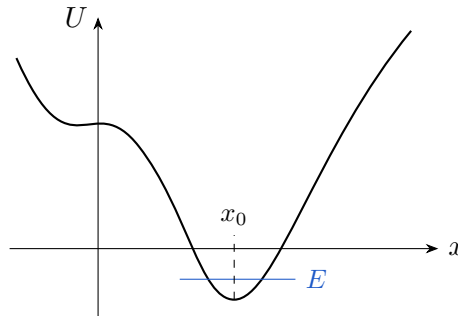


Figure 1.2: Expansion around a minimum x_0 of the potential.

For $H = p^2/2m + U(x)$, with $x = x_0 + q$,

$$U(q) = U(x_0) + \frac{1}{2}U''(x_0)q^2 + \frac{1}{3!}U'''(x_0)q^3 + \dots$$

The condition for the harmonic approximation is $q \ll |U''/U'''|_{x=x_0}$ (the higher terms might otherwise change the physics).

In several dimensions, $\vec{x} = \vec{x}_0 + \vec{q}$,

$$U(\vec{x}) = U(\vec{x}_0) + \frac{1}{2} \sum_{i,j} \underbrace{\frac{\partial^2 U}{\partial x_i \partial x_j}}_{\text{Hessian, rigidity}} q_i q_j + \dots$$

Notation. For one degree of freedom,

$$H = \frac{p^2}{2m} + \frac{\kappa}{2}q^2 = -\frac{\hbar^2}{2m} \frac{d^2}{dq^2} + \frac{\kappa}{2}q^2.$$

Balancing the two terms, $\frac{\hbar^2}{2m} \frac{1}{\ell^2} = \frac{\kappa \ell^2}{2}$, defines the **oscillator length** and energy

$$\ell^2 = \sqrt{\frac{\hbar^2}{\kappa m}}, \quad E_0 = \kappa \ell^2 = \hbar \omega = \hbar \sqrt{\frac{\kappa}{m}}.$$

With $E = E_0 \varepsilon$, $q = \ell x$,

$$H = \hbar \omega \left(-\frac{1}{2} \frac{d^2}{dx^2} + \frac{x^2}{2} \right) \implies -\frac{1}{2} \psi'' + \frac{x^2}{2} \psi = \varepsilon \psi.$$

Wigner–Heisenberg algebra.

$$a = \frac{1}{\sqrt{2}} \left(x + \frac{d}{dx} \right), \quad a^\dagger = \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx} \right), \quad N = a^\dagger a, \quad [a, a^\dagger] = 1.$$

Given the W–H algebra:

1. $\exists |0\rangle \in \mathcal{H}$ such that $a|0\rangle = 0$;
2. $N|n\rangle = n|n\rangle$, $n \in \mathbb{Z}_{\geq 0}$;
3. $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$ and $a|n\rangle = \sqrt{n}|n-1\rangle$.

For the harmonic oscillator,

$$H = \hbar \omega \left(a^\dagger a + \frac{1}{2} \right).$$

Vacuum state. $a|0\rangle = 0 \iff (x + \frac{d}{dx})\psi_0 = 0$, so

$$\psi_0(x) = \frac{1}{\pi^{1/4}} e^{-x^2/2}, \quad \psi_n(x) = \frac{1}{\sqrt{n!}} \left[\frac{1}{\sqrt{2}} \left(x - \frac{d}{dx} \right) \right]^n \psi_0(x)$$

(Hermite polynomials times the Gaussian).

Indeed, $[N, a] = [a^\dagger a, a] = [a^\dagger, a]a = -a$, so if $N|\lambda\rangle = \lambda|\lambda\rangle$,

$$N(a|\lambda\rangle) = aN|\lambda\rangle - a|\lambda\rangle = (\lambda - 1)(a|\lambda\rangle), \quad N(a^\dagger|\lambda\rangle) = (\lambda + 1)(a^\dagger|\lambda\rangle),$$

i.e. $a|\lambda\rangle \propto |\lambda - 1\rangle$. Since $\|a|\lambda\rangle\|^2 = \lambda \geq 0$, the lowering must terminate at a state with $a|\lambda_0\rangle = 0$.

1.2 Extended Wigner–Heisenberg algebra

$$H = H_1 + H_2 + \cdots + H_K, \quad [H_i, H_j] = 0 \quad \Longrightarrow \quad E = \sum_{i=1}^K E_i,$$

$$H = \sum_{i=1}^K \hbar \omega_i \left(a_i^\dagger a_i + \frac{1}{2} \right), \quad [a_i, a_j^\dagger] = \delta_{ij}.$$

In this case the vacuum state is

$$|0\rangle = |0\rangle_1 \otimes |0\rangle_2 \otimes \cdots \otimes |0\rangle_K \iff \Psi_0(\vec{x}_1, \dots, \vec{x}_K) = \psi_0(\vec{x}_1) \cdots \psi_0(\vec{x}_K),$$

- $a_i |0\rangle = 0$ for all i (a_i : annihilation operators);
- $|\lambda\rangle = |n_1\rangle \otimes |n_2\rangle \otimes \cdots \otimes |n_K\rangle = C (a_1^\dagger)^{n_1} (a_2^\dagger)^{n_2} \cdots (a_K^\dagger)^{n_K} |0\rangle$ ($a_i^\dagger$: creation operators), where $\lambda = \{n_1, n_2, \dots, n_K\}$ are the **occupation numbers**;
- $H |\lambda\rangle = E_\lambda |\lambda\rangle$ with

$$E_\lambda = \sum_{i=1}^K \frac{\hbar \omega_i}{2} + \sum_{i=1}^K n_i \hbar \omega_i.$$

Generic case.

$$H = \sum_{i=1}^K \frac{p_i^2}{2m_i} + \frac{1}{2} \sum_{i,j} q_i K_{ij} q_j.$$

The canonical transformation $p_i = \sqrt{m_i} \pi_i$, $q_i = \xi_i / \sqrt{m_i}$ (which keeps $[q_i, p_i] = i\hbar$) gives

$$H = \sum_{i=1}^K \frac{\pi_i^2}{2} + \frac{1}{2} \sum_{i,j} \xi_i \Omega_{ij} \xi_j, \quad \Omega_{ij} = \frac{K_{ij}}{\sqrt{m_i m_j}} \sim \omega^2.$$

An orthogonal transformation diagonalises Ω :

$$\Lambda^\top \Omega \Lambda = \text{diag}(\omega_1^2, \dots, \omega_K^2), \quad \Lambda^\top \Lambda = \Lambda \Lambda^\top = \mathbb{I},$$

and with $\vec{P} = \Lambda^\top \vec{\pi}$, $\vec{Q} = \Lambda^\top \vec{\xi}$ (orthogonal transformations preserve the commutation relations of the original variables, so this is again a canonical transformation),

$$H = \sum_{i=1}^K \left(\frac{P_i^2}{2} + \frac{\omega_i^2}{2} Q_i^2 \right)$$

— the **normal modes**.

1.3 Exercises

1.2 (a) Derive $[a, a^\dagger] = 1$, $[N, a] = -a$, $[N, a^\dagger] = a^\dagger$ for $a = \frac{1}{\sqrt{2}}(x + d/dx)$, $a^\dagger = \frac{1}{\sqrt{2}}(x - d/dx)$, $N = a^\dagger a$.

$$\begin{aligned} [a, a^\dagger]f &= \frac{1}{2} \left[x + \frac{d}{dx}, x - \frac{d}{dx} \right] f = \frac{1}{2} \left(- \left[x, \frac{d}{dx} \right] + \left[\frac{d}{dx}, x \right] \right) f = \left[\frac{d}{dx}, x \right] f \\ &= \frac{d}{dx}(xf) - x \frac{df}{dx} = f \implies [a, a^\dagger] = 1; \\ [N, a] &= [a^\dagger a, a] = a^\dagger \underbrace{[a, a]}_0 + \underbrace{[a^\dagger, a]}_{-1} a = -a; \\ [N, a^\dagger] &= ([a, N])^\dagger = (a)^\dagger = a^\dagger. \end{aligned}$$

(b) Show that a and $a^\dagger$ are Hermitian conjugates, i.e. $\langle n|a|m\rangle^* = \langle m|a^\dagger|n\rangle$:

$$\begin{aligned}\langle n|a|m\rangle^* &= \frac{1}{\sqrt{2}} \int dx \psi_n(x) \left(x + \frac{d}{dx} \right) \psi_m^*(x) \\ &= \frac{1}{\sqrt{2}} \int dx \left(\psi_m^* x \psi_n + \underbrace{\frac{d}{dx}(\psi_n \psi_m^*)}_{\text{boundary term}} - \psi_m^* \frac{d}{dx} \psi_n \right) = \frac{1}{\sqrt{2}} \int dx \psi_m^*(x) \left(x - \frac{d}{dx} \right) \psi_n(x),\end{aligned}$$

so $\langle n|a|m\rangle^* = \langle m|\frac{1}{\sqrt{2}}(x - d/dx)|n\rangle$ and $a^\dagger = \frac{1}{\sqrt{2}}(x - d/dx)$.

(c) Show that $H = \hbar\omega(N + \frac{1}{2})$:

$$H = \frac{\hbar\omega}{2} \left(-\frac{d^2}{dx^2} + x^2 \right), \quad \left(x - \frac{d}{dx} \right) \left(x + \frac{d}{dx} \right) f = \left(-\frac{d^2}{dx^2} + x^2 \right) f - f,$$

so $-d^2/dx^2 + x^2 = 2a^\dagger a + 1$ and $H = \hbar\omega(N + \frac{1}{2})$.

1.3 (a) From $[a, a^\dagger] = 1$ derive $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$, where $N|n\rangle = n|n\rangle$:

$$N(a^\dagger|n\rangle) = ([N, a^\dagger] + a^\dagger N)|n\rangle = a^\dagger(1 + N)|n\rangle = (1 + n)(a^\dagger|n\rangle),$$

so $a^\dagger|n\rangle = c|n+1\rangle$, and $|c|^2 = \langle n|aa^\dagger|n\rangle = \langle n|[a, a^\dagger] + a^\dagger a|n\rangle = 1 + n$, i.e. $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$.

(b) Normalise $|nm\rangle = C_{nm}(a_1^\dagger)^n(a_2^\dagger)^m|\text{vac}\rangle$, with $[a_i^\dagger, a_j^\dagger] = [a_i, a_j] = 0$, $[a_i, a_j^\dagger] = \delta_{ij}$:

$$\langle nm|nm\rangle = |C_{nm}|^2 \langle 0|a_1^n a_2^m (a_1^\dagger)^n (a_2^\dagger)^m|0\rangle = |C_{nm}|^2 \langle 0|a_1^n (a_1^\dagger)^n|0\rangle \langle 0|a_2^m (a_2^\dagger)^m|0\rangle = |C_{nm}|^2 A_n A_m,$$

with $A_n = \langle 0|a^n (a^\dagger)^n|0\rangle$. Moving one a to the right through the n creation operators, $a(a^\dagger)^n = n(a^\dagger)^{n-1} + (a^\dagger)^n a$, so

$$A_n = n A_{n-1} + \langle 0|a^{n-1}(a^\dagger)^n a|0\rangle = n A_{n-1} \implies A_n = n! A_0 = n!$$

(since $A_0 = \langle 0|0\rangle = 1$). Hence $|C_{nm}|^2 m! n! = 1$ and

$$|C_{nm}| = \frac{1}{\sqrt{n! m!}}.$$

2 Dielectric crystals and phonons

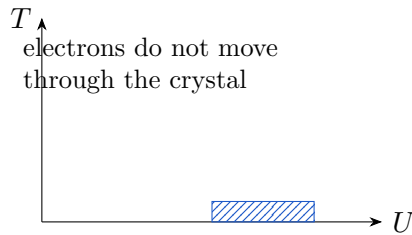


Figure 2.1: Dielectric crystals: strong interactions, crystalline order, no mobile electrons.

The task is to understand such systems from a quantum point of view, as $T \rightarrow 0$: the ground state, the low-energy excitations and the physical properties.

2.1 One-dimensional monoatomic crystal

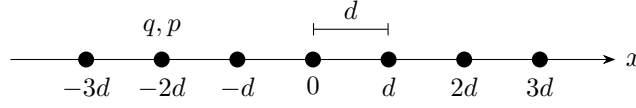


Figure 2.2: The lattice $\mathcal{L} = \{0, \pm d, \pm 2d, \dots\}$; d is the period of the lattice. (Example: ultracold systems.)

With $q(x), p(x)$ for $x \in \mathcal{L}$,

$$H = \sum_{x \in \mathcal{L}} \frac{p^2(x)}{2m} + \frac{1}{2} \sum_{x, y \in \mathcal{L}} q(x) M(x, y) q(y),$$

where m is the same for every site because the crystal is monoatomic.

Symmetries of the system.

- Translation I: $M(x + d, y + d) = M(x, y)$, hence $M(x, y) = M(x - y)$.
- Reflection: $M(x) = M(-x)$.
- Translation II: a displacement of *all* atoms by the same distance q costs no energy,

$$\sum_{x, y} M(x - y) (q(x) + q) (q(y) + q) = \sum_{x, y} M(x - y) q(x) q(y) \implies \sum_{x \in \mathcal{L}} M(x) = 0.$$

Physical constraint: $M(x)$ has finite range R .

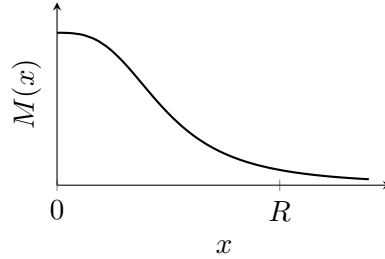


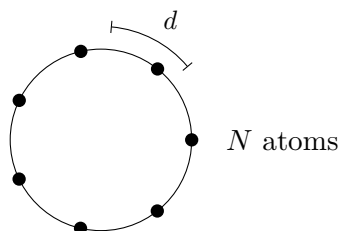
Figure 2.3: The coupling $M(x)$ decays beyond a range R .

2.2 Lattice Fourier transform

Impose the **Born–von Kármán** (periodic) boundary condition:

$$\mathcal{H} = \{ f : \mathcal{L} \rightarrow \mathbb{C} \mid f(x + L) = f(x) \}, \quad L = Nd, \quad N \in \mathbb{Z} \text{ (large)}.$$

Effectively the N atoms sit on a ring, and the sites form $\mathcal{L}_N = \mathcal{L}/N\mathcal{L}$ (x and $x + L$ are the same site).



Inner product.

$$(f, g) := \sum_{x \in \mathcal{L}_N} f^*(x)g(x)$$

— a finite sum, thanks to the boundary conditions.

Translation operator. $(Tf)(x) = f(x + d)$ (elementary translation). Using the definition of the inner product, $T^\dagger = T^{-1}$. Since T is unitary its eigenvalues are phases, $Te(x) = e^{i\gamma}e(x)$, i.e. $e(x + d) = e^{i\gamma}e(x)$, so $e(x) = e^{ikx}$ with $e^{i\gamma} = e^{ikd}$.

The normalised eigenfunctions

$$e_k(x) = \frac{1}{\sqrt{N}} e^{ikx}$$

generate a basis of the (N -dimensional) Hilbert space. There seem to be infinitely many e_k ; however

- the boundary condition $e_k(x + Nd) = e_k(x)$ requires $Nkd = 2\pi m$, i.e. $k = 2\pi m/L$, $m \in \mathbb{Z}$ — still infinitely many;
- $e_{k+k'}(x) = e_k(x)$ for all $x \in \mathcal{L}$ when $e^{ik'x} = 1$, i.e. $k' = 2\pi n/d$.

Hence

$$k = \frac{2\pi}{L}m, \quad m \in \mathbb{Z}/N\mathbb{Z},$$

and these N values of k form the **Brillouin zone** $\mathcal{B}$.

Fourier transform. For $f \in \mathcal{H}$ the direct transform $\tilde{f} : \mathcal{B} \rightarrow \mathbb{C}$ and its inverse are

$$\tilde{f}_k := \sum_{x \in \mathcal{L}_N} e^{-ikx} f(x) = \sqrt{N} (e_k, f), \quad f(x) = \sum_{k \in \mathcal{B}} (e_k, f) e_k(x) = \frac{1}{N} \sum_{k \in \mathcal{B}} \tilde{f}_k e^{ikx}.$$

Parseval identity:

$$\sum_{x \in \mathcal{L}_N} f^*(x)g(x) = \frac{1}{N} \sum_{k \in \mathcal{B}} \tilde{f}_k^* \tilde{g}_k.$$

Continuum limit ($d \rightarrow 0$, $N \rightarrow \infty$). Defining $\tilde{f}(k) = d \tilde{f}_k = d \sum_x e^{-ikx} f(x) \rightarrow \int dx f(x) e^{-ikx}$ and using $\Delta k = 2\pi/(Nd)$,

$$\tilde{f}(k) = \int_{-\infty}^{\infty} dx f(x) e^{-ikx}, \quad f(x) = \int_{-\infty}^{\infty} \frac{dk}{2\pi} \tilde{f}(k) e^{ikx}.$$

2.3 Normal modes of the monoatomic crystal

We need the eigenvalue problem

$$\sum_{y \in \mathcal{L}_N} M(x - y) \phi(y) = \kappa \phi(x), \quad \text{i.e.} \quad M\phi = \kappa \phi.$$

Since $[M, T] = 0$, M and T have the same eigenvectors; in Fourier space $Me_k = \kappa_k e_k$, with

$$\kappa_k = \sum_{y \in \mathcal{L}_N} M(y) e^{-iky}.$$

With $\omega_k = \sqrt{\kappa_k/m}$,

$$H = \sum_{k \in \mathcal{B}} \hbar \omega_k \left(a_k^\dagger a_k + \frac{1}{2} \right).$$

The local variables in terms of creation and annihilation operators are

$$q(x) = \sum_{k \in \mathcal{B}} \sqrt{\frac{\hbar}{2mN\omega_k}} [a_k e^{ikx} + a_k^\dagger e^{-ikx}],$$

$$p(x) = -i \sum_{k \in \mathcal{B}} \sqrt{\frac{\hbar m \omega_k}{2N}} [a_k e^{ikx} - a_k^\dagger e^{-ikx}].$$

Translation II gives $\kappa_{k=0} = \sum_x M(x) = 0$, hence $\omega_0 = 0$. Since M has finite range, $\kappa_k = \sum_{|x| < R} M(x) e^{-ikx}$ is an analytic function of k , even in k , and near $k = 0$

$$\kappa_k = 0 + \frac{\kappa_0''}{2} k^2 + \dots \implies \omega_k \simeq v_s |k|, \quad v_s = \sqrt{\frac{\kappa_0''}{2m}}.$$

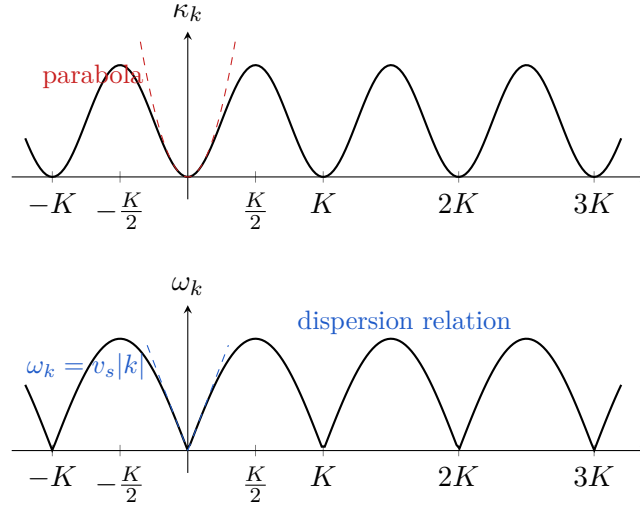


Figure 2.4: κ_k is periodic with period $K = 2\pi/d$ and parabolic near its zeros; the phonon dispersion ω_k is linear (gapless) near $k = 0$.

The ground state of the system breaks the (continuous translation) symmetry of space; the gapless linear branch is a consequence of the **Goldstone theorem**.

Definition of the vacuum.

$$|\text{vac}\rangle = |0\rangle, \quad a_k |0\rangle = 0 \quad \forall k \in \mathcal{B}.$$

$a_k^\dagger |0\rangle = |k\rangle$ is an **elementary excitation**, the **phonon**, with quasi-momentum $\hbar k$ and energy $E_k = \hbar \omega_k$. In general

$$|\lambda\rangle = \mathcal{N} (a_{k_1}^\dagger)^{n_1} \dots (a_{k_N}^\dagger)^{n_N} |0\rangle, \quad \lambda = \{n_1, \dots, n_N\},$$

with $\mathcal{N}$ a normalisation constant, and

$$E_\lambda = \sum_{k \in \mathcal{B}} \hbar \omega_k \left(n_k + \frac{1}{2} \right).$$

2.4 Exercises

2.2 (a) With $(Tf)(x) = f(x+d)$ and $(f, g) = \sum_{x \in \mathcal{L}_N} f^*(x)g(x)$,

$$(f, f) = (Tf, Tf) = \sum_{x \in \mathcal{L}_N} (Tf)^*(Tf) = (f, T^\dagger Tf) \implies T^\dagger T = \mathbb{K}.$$

(b) With $e_k = e^{ikx}/\sqrt{N}$, $k \in \frac{2\pi}{L}\mathbb{Z}$,

$$(e_k, e_p) = \sum_{x \in \mathcal{L}_N} \frac{1}{\sqrt{N}} e^{-ikx} \frac{1}{\sqrt{N}} e^{ipx} = \frac{1}{N} \underbrace{\sum_{x \in \mathcal{L}_N} e^{-i(k-p)x}}_{N\delta_{kp}} = \delta_{kp}.$$

(c) $e_k(x+L) = \frac{1}{\sqrt{N}} e^{ik(x+L)} = e_k(x) e^{ikL}$ and $kL \in 2\pi\mathbb{Z}$, so $e_k(x+L) = e_k(x)$.

2.6 (a) Write

$$q(x) = \frac{1}{\sqrt{N}} \sum_{k \in \mathcal{B}} \ell_k Q_k e^{ikx}, \quad p(x) = \frac{1}{\sqrt{N}} \sum_{k \in \mathcal{B}} \frac{\hbar}{\ell_k} P_k e^{ikx}, \quad \ell_k^2 = \frac{\hbar}{m\omega_k}, \quad \omega_k = \sqrt{\frac{\kappa_k}{m}}.$$

Hermiticity, $q^\dagger(x) = q(x)$ and $p^\dagger(x) = p(x)$, gives $Q_k^\dagger = Q_{-k}$, $P_k^\dagger = P_{-k}$. Then

$$\sum_{x \in \mathcal{L}_N} \frac{p^2(x)}{2m} = \frac{\hbar^2}{2mN} \sum_{k,p \in \mathcal{B}} \frac{P_k P_p}{\ell_k \ell_p} \underbrace{\sum_x e^{i(k+p)x}}_{N\delta_{k,-p}} = \sum_{k \in \mathcal{B}} \frac{\hbar^2}{2m\ell_k^2} P_k P_{-k},$$

$$\begin{aligned} \sum_{x,y \in \mathcal{L}_N} q(x) M(x-y) q(y) &= \frac{1}{N} \sum_{k,p \in \mathcal{B}} \ell_k \ell_p Q_k Q_p \sum_x e^{i(k+p)x} \sum_y M(y-x) e^{ip(y-x)} \\ &= \frac{1}{N} \sum_{k,p} \ell_k \ell_p Q_k Q_p N\delta_{k+p,0} \kappa_p = \sum_{k \in \mathcal{B}} \ell_k^2 \kappa_k Q_k Q_{-k}. \end{aligned}$$

Since $\frac{\hbar^2}{2m\ell_k^2} = \frac{\hbar\omega_k}{2}$ and $\frac{1}{2}\ell_k^2 \kappa_k = \frac{\hbar\omega_k}{2}$,

$$H = \sum_{k \in \mathcal{B}} \frac{\hbar\omega_k}{2} (P_k P_{-k} + Q_k Q_{-k}).$$

(b) With $a_k = \frac{1}{\sqrt{2}}(Q_k + iP_k)$ and $a_k^\dagger = \frac{1}{\sqrt{2}}(Q_{-k} - iP_{-k})$,

$$\begin{aligned} H &= \sum_{k \in \mathcal{B}} \frac{\hbar\omega_k}{2} [(Q_{-k} - iP_{-k})(Q_k + iP_k) - i(Q_k P_{-k} - P_k Q_{-k})] \\ &= \sum_{k \in \mathcal{B}} \hbar\omega_k \left[a_k^\dagger a_k - \frac{i}{2}(Q_k P_{-k} - P_k Q_{-k}) \right] \end{aligned}$$

(using $\omega_k = \omega_{-k}$ to relabel $k \rightarrow -k$ in the cross terms). Inverting the expansions,

$$Q_k = \frac{1}{\ell_k \sqrt{N}} \sum_x q(x) e^{-ikx}, \quad P_k = \frac{\ell_k}{\hbar \sqrt{N}} \sum_x p(x) e^{-ikx},$$

and using $[q(x), p(y)] = i\hbar\delta_{xy}$,

$$\begin{aligned} Q_k P_{-k} &= \frac{1}{\hbar N} \sum_{x,y} q(x) p(y) e^{ik(y-x)}, \\ P_k Q_{-k} &= \frac{1}{\hbar N} \sum_{x,y} p(y) q(x) e^{ik(x-y)} = \frac{1}{\hbar N} \sum_{x,y} (q(x) p(y) - i\hbar\delta_{xy}) e^{ik(x-y)}, \\ Q_k P_{-k} - P_k Q_{-k} &= i + \frac{1}{\hbar N} \sum_{x,y} q(x) p(y) (e^{ik(y-x)} - e^{ik(x-y)}). \end{aligned}$$

The last term disappears when summed over k (with the even weight ω_k), because it is odd under $k \rightarrow -k$. Therefore $-\frac{i}{2} \cdot i = \frac{1}{2}$ and

$$H = \sum_{k \in \mathcal{B}} \hbar \omega_k \left(a_k^\dagger a_k + \frac{1}{2} \right).$$

2.7 (a) With $q(x) = \sum_{p \in \mathcal{B}} \sqrt{\frac{\hbar}{2Nm\omega_p}} (a_p e^{ipx} + a_p^\dagger e^{-ipx})$ and, for a fixed k , the normalised state $|\psi\rangle = \frac{1}{\sqrt{2}} (a_k^\dagger |0\rangle + |0\rangle)$, compute $\langle\psi|q(x)|\psi\rangle$. Only the $p = k$ term connects $|0\rangle$ and $|k\rangle$:

$$\begin{aligned} (a_k e^{ikx} + a_k^\dagger e^{-ikx}) (a_k^\dagger |0\rangle + |0\rangle) &= e^{ikx} |0\rangle + e^{-ikx} a_k^\dagger |0\rangle + (\text{two-phonon state}), \\ (\langle 0| + \langle 0| a_k) (e^{ikx} |0\rangle + e^{-ikx} a_k^\dagger |0\rangle) &= e^{ikx} + e^{-ikx} = 2 \cos(kx), \end{aligned}$$

so

$$\langle\psi|q(x)|\psi\rangle = \frac{1}{2} \sqrt{\frac{\hbar}{2Nm\omega_k}} 2 \cos(kx) = \sqrt{\frac{\hbar}{2Nm\omega_k}} \cos(kx).$$

3 Thermodynamics and linear response of the harmonic crystal

Questions one can always ask about a system S :

- **Thermodynamics**: the (static) system in contact with a bath at temperature $T \rightarrow$ response functions, entropy, ...
- **Dynamics**: the system under time-dependent perturbations $A(t)$; the response of an observable is $B(t) = \int_{t_0}^t K(t-t') A(t') dt'$.
- **Kinetics**: dissipation, equilibration, transport, ... (temperature is not a perturbation). This requires the exchange of quantum numbers; in our case the phonons do not interact, so it does not apply here.

3.1 Thermodynamics

For a quantum system with $H(\xi) |\lambda\rangle = E_\lambda(\xi) |\lambda\rangle$ in contact with a bath at temperature T (we set $k_B = 1$), the **Gibbs measure** is

$$w_\lambda = \frac{1}{Z} e^{-E_\lambda/T}, \quad Z = \sum_\lambda e^{-E_\lambda(\xi)/T},$$

where Z is the **partition function**: all the thermodynamic information is contained in it. The **Helmholtz free energy** is

$$Z = e^{-F/T} \iff F = -T \ln Z.$$

The mean energy and heat capacity are

$$E := \sum_{\lambda} w_{\lambda} E_{\lambda} = \frac{T^2}{Z} \frac{\partial Z}{\partial T} \Big|_{\xi} = -T^2 \frac{\partial}{\partial T} \left(\frac{F}{T} \right) \Big|_{\xi}, \quad C := \frac{\partial E}{\partial T} = -T \frac{\partial^2 F}{\partial T^2} \Big|_{\xi}.$$

In our case there is no distinction between C_V and C_P .

Remark. For two independent subsystems in contact with the same bath, $E = E_1 + E_2$, so $Z = Z_1 Z_2$ and $F = F_1 + F_2$.

Normal modes of a harmonic system. With $H = \sum_{k \in \mathcal{B}} \hbar \omega_k (N_k + \frac{1}{2})$, $N_k = a_k^{\dagger} a_k$,

$$Z = \sum_{\lambda} e^{-\sum_k \hbar \omega_k (n_k + \frac{1}{2})/T} = \prod_{k \in \mathcal{B}} \sum_{n=0}^{\infty} e^{-\hbar \omega_k n/T} e^{-\hbar \omega_k /2T} = e^{-\frac{\hbar}{2T} \sum_k \omega_k} \prod_{k \in \mathcal{B}} \frac{1}{1 - e^{-\hbar \omega_k /T}},$$

$$F = \frac{\hbar}{2} \sum_{k \in \mathcal{B}} \omega_k + T \sum_{k \in \mathcal{B}} \ln(1 - e^{-\hbar \omega_k /T}).$$

Condensed matter means a large number of particles, so $\sum_{k \in \mathcal{B}} f(k) = \frac{L}{2\pi} \sum_k \Delta k f(k) \rightarrow L \int_{\mathcal{B}} \frac{dk}{2\pi} f(k)$ and

$$F = \frac{L\hbar}{2} \int_{\mathcal{B}} \frac{dk}{2\pi} \omega_k + TL \int_{\mathcal{B}} \frac{dk}{2\pi} \ln(1 - e^{-\hbar \omega_k /T}).$$

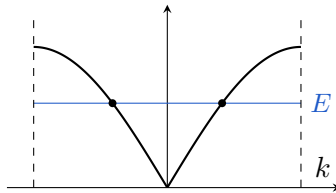
For any function of the energy,

$$\int \frac{dk}{2\pi} f(\hbar \omega_k) = \int \frac{dk}{2\pi} \int dE \delta(E - \hbar \omega_k) f(E) = \int dE f(E) \rho(E),$$

with the **local density of states** ($L\rho(E)$ is the density of states)

$$\rho(E) = \int_{\mathcal{B}} \frac{dk}{2\pi} \delta(E - \hbar \omega_k) = \frac{1}{2\pi} \cdot 2 \cdot \frac{1}{|\hbar \omega'_k|} \Big|_{\hbar \omega_k = E},$$

the factor 2 coming from the two solutions $\pm k$ of $\hbar \omega_k = E$ in the Brillouin zone.



Hence

$$F = \frac{L}{2} \int dE \rho(E) E + LT \int dE \rho(E) \ln(1 - e^{-E/T}).$$

The details of the particular system are hidden in the local density of states $\rho(E)$. The heat capacity is

$$C_V = -T \frac{\partial^2 F}{\partial T^2} = L \int dE \rho(E) \underbrace{\frac{E^2}{(2T)^2 \sinh^2(E/2T)}}_{f(E/T) \text{ (universal part)}},$$

where $f(E/T)$ is a bell-shaped function of width $\sim T$, with $f(0) = 1$.

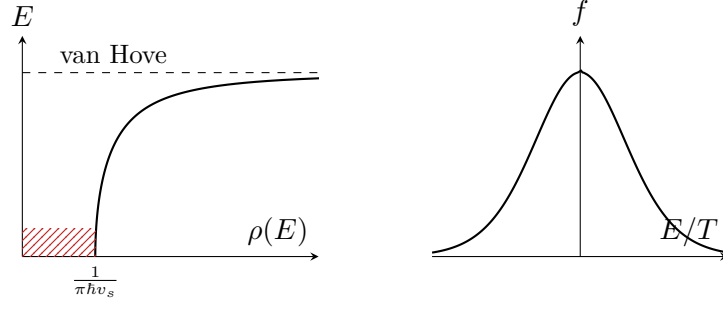


Figure 3.1: Left: local density of states of the monoatomic chain (constant at low energy, van Hove singularity at the top of the band). At low T only the shaded region contributes. Right: the universal function $f(E/T)$.

As $T \rightarrow 0$ only $E \lesssim T$ contributes, where $\rho(E) \simeq 1/(\pi\hbar v_s)$; with $z = E/T$ and $\int_0^\infty dz \, z^2/(4 \sinh^2(z/2)) = \pi^2/3$,

$$C_V = \frac{\pi T L}{3\hbar v_s} \quad (T \rightarrow 0).$$

3.2 Dynamics: linear response and the Kubo formula

Let A and B be observables of a system in equilibrium, and couple A to an external signal $\phi(t)$:

$$H = H_0 + A\phi(t) \implies \langle B(t) \rangle = ?$$

With $|t\rangle = \mathcal{U}(t, 0)|0\rangle$, $\langle B(t) \rangle = \langle 0|\mathcal{U}^\dagger(t, 0)B\mathcal{U}(t, 0)|0\rangle$, and

$$i\hbar\partial_t\mathcal{U} = (H_0 + A\phi(t))\mathcal{U}, \quad \mathcal{U} = \mathcal{U}_0\tilde{\mathcal{U}}, \quad \mathcal{U}_0 = e^{-iH_0t/\hbar}.$$

Since $i\hbar(\partial_t\mathcal{U}_0)\tilde{\mathcal{U}} = H_0\mathcal{U}_0\tilde{\mathcal{U}}$,

$$i\hbar\partial_t\tilde{\mathcal{U}} = \tilde{A}(t)\phi(t)\tilde{\mathcal{U}}, \quad \tilde{A} = \mathcal{U}_0^{-1}A\mathcal{U}_0.$$

Writing $\tilde{\mathcal{U}} = \mathbb{K} + \tilde{\mathcal{U}}^{(1)} + \dots$,

$$i\hbar\partial_t\tilde{\mathcal{U}}^{(1)} = \tilde{A}\phi \implies \tilde{\mathcal{U}}(t, 0) \simeq \mathbb{K} - \frac{i}{\hbar} \int_0^t dt' \tilde{A}(t')\phi(t'),$$

the first terms of $\tilde{\mathcal{U}}(t, 0) = \mathcal{T} \exp(-\frac{i}{\hbar} \int_0^t dt' \tilde{A}(t')\phi(t'))$ ($\mathcal{T}$: time ordering). Then

$$B(t) := \langle B \rangle = \langle B \rangle_0 - \frac{i}{\hbar} \left\langle \left[\tilde{B}(t), \int_0^t dt' \tilde{A}(t')\phi(t') \right] \right\rangle.$$

Let us assume $\langle B \rangle_0 = 0$ and let the perturbation start at $t_0 \rightarrow -\infty$:

$$B(t) = -\frac{i}{\hbar} \int_{-\infty}^{+\infty} \underbrace{\left\langle [\tilde{B}(t), \tilde{A}(t')] \right\rangle}_{\text{correlation function}} \theta(t-t') \phi(t') dt'.$$

The correlation function involves only observables of the system (internal to it); ϕ describes the external apparatus; the factor $\theta(t-t')$ gives the causal structure (only the past matters). The **retarded Green function** (propagator) is

$$G^R(t-t') = -\frac{i}{\hbar} \left\langle [\tilde{B}(t), \tilde{A}(t')] \right\rangle \theta(t-t'), \quad B(t) = \int G^R(t-t')\phi(t') dt'.$$

Example. Apply a force $F(t)$ to the atom at x' of the chain: $\delta H = -F(t) q(x')$, so $A = -q(x')$. What is $p(x, t)$?

$$p(x, t) = \int_{-\infty}^{\infty} \left(\frac{i}{\hbar} \langle [\tilde{p}(x, t), \tilde{q}(x', t')] \rangle \theta(t - t') \right) F(t') dt',$$

which defines the **retarded propagator** $\Delta^R(x, x'; t, t') = \Delta^R(x - x', t - t')$ (by symmetry). In the interaction picture,

$$\tilde{a}_k = e^{iH_0 t/\hbar} a_k e^{-iH_0 t/\hbar}, \quad \frac{d\tilde{a}_k}{dt} = \frac{i}{\hbar} [H_0, \tilde{a}_k] = -i\omega_k \tilde{a}_k \implies \tilde{a}_k = a_k e^{-i\omega_k t}, \quad \tilde{a}_k^\dagger = a_k^\dagger e^{i\omega_k t},$$

$$\tilde{q}(x, t) = \sum_k \sqrt{\frac{\hbar}{2mN\omega_k}} [a_k e^{ikx - i\omega_k t} + \text{h.c.}],$$

$$\tilde{p}(x, t) = -i \sum_k \sqrt{\frac{\hbar m \omega_k}{2N}} [a_k e^{ikx - i\omega_k t} - \text{h.c.}],$$

$$[\tilde{p}(x, t), \tilde{q}(x', t')] = \frac{\hbar(-i)}{2N} \sum_{k \in \mathcal{B}} [e^{ik(x-x') - i\omega_k(t-t')} + \text{c.c.}].$$

The commutator is a c -number (it contains no operators), so the state used for the average does not matter. Therefore

$$\Delta^R(x - x', t - t') = \frac{1}{N} \sum_{k \in \mathcal{B}} \cos[(x - x')k - (t - t')\omega_k] \theta(t - t').$$

For a short pulse at $x' = 0, t' = 0$, $F(t') = F\Delta t \delta(t')$:

$$p(x, t) = \frac{1}{N} \sum_{k \in \mathcal{B}} \cos(xk - t\omega_k) F\Delta t \simeq F\Delta t d \int_{\mathcal{B}} \frac{dk}{2\pi} \cos\left(x\left(k - \omega_k \frac{t}{x}\right)\right)$$

(t/x is the inverse of a speed). For large x , by the **stationary phase approximation**, this tends to zero unless

$$\frac{d}{dk} \left(k - \omega_k \frac{t}{x} \right) = 0 \iff \omega'_k = \frac{x}{t}.$$

Since the group velocity ω'_k is bounded (maximal, $= v_s$, at $k = 0$), the signal is non-zero only inside a “light cone” $|x| \lesssim v_s t$; information cannot travel outside it.

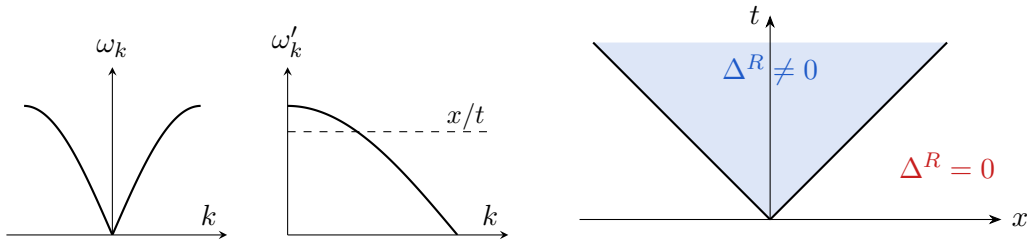


Figure 3.2: Dispersion relation and group velocity; the retarded propagator vanishes outside the cone $|x| > v_s t$.

4 Polyatomic and three-dimensional lattices

4.1 One-dimensional diatomic crystal

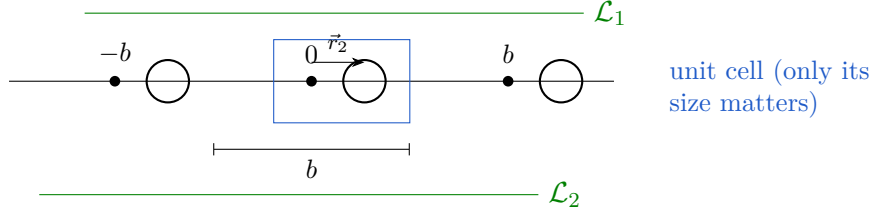


Figure 4.1: Diatomic chain: two sublattices $\mathcal{L}_1, \mathcal{L}_2$ and the unit cell.

With the **Bravais lattice** $\mathcal{L} = b\mathbb{Z}$, the sublattices are $\mathcal{L}_i = \mathcal{L} + r_i$; the set $\mathcal{B} = \{r_i\}$ is called the **basis**. The physical quantities are $q(x), p(x)$ for $x \in \mathcal{L}_1 \cup \mathcal{L}_2$; we write $q_\alpha(x) = q(x + r_\alpha)$ with $r_\alpha \in \mathcal{B}, x \in \mathcal{L}$:

$$H = \sum_{x \in \mathcal{L}} \sum_{\alpha=1}^2 \frac{p_\alpha^2(x)}{2m_\alpha} + \frac{1}{2} \sum_{x,y \in \mathcal{L}} \sum_{\alpha,\beta} q_\alpha(x) M_{\alpha\beta}(x-y) q_\beta(y).$$

Due to symmetry (moving the whole lattice by q),

$$\sum_{\alpha\beta} \sum_{x,y} (q_\alpha(x) + q) M_{\alpha\beta}(x-y) (q_\beta(y) + q) = \sum_{\alpha\beta} \sum_{x,y} q_\alpha(x) M_{\alpha\beta}(x-y) q_\beta(y),$$

hence

$$\sum_{\beta=1}^2 \sum_{x \in \mathcal{L}} M_{\alpha\beta}(x) = 0 \iff \sum_{\beta=1}^2 \tilde{M}_{\alpha\beta}(0) = 0.$$

This property predicts a zero frequency at $k = 0$.

Normal modes. With $K_{\alpha\beta}(x-y) := M_{\alpha\beta}(x-y)/\sqrt{m_\alpha m_\beta}$,

$$\sum_{y,\beta} K_{\alpha\beta}(x-y) \phi_\beta(y) = \omega^2 \phi_\alpha(x), \quad \phi_\beta(y) = e^{ik(y+r_\beta)} \tilde{\phi}_\beta(k).$$

After the Fourier transformation,

$$\tilde{K}_{\alpha\beta}(k) \tilde{\phi}_\beta(k) = \omega^2 \tilde{\phi}_\alpha(k), \quad \tilde{K}_{\alpha\beta}(k) := \sum_{x \in \mathcal{L}} \frac{M_{\alpha\beta}(x)}{\sqrt{m_\alpha m_\beta}} e^{ik(r_\beta - r_\alpha) - ikx},$$

a 2×2 matrix with two eigenvalues $\omega_i^2 = \omega_i^2(k)$, $i \in \{1, 2\}$. As before, $\phi_\beta(y) = \phi_\beta(y + L)$ with period $L = Nb$.

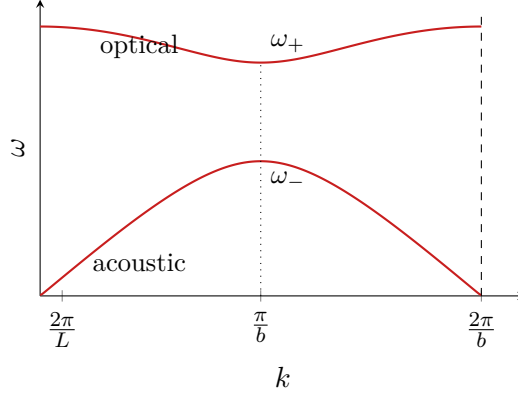


Figure 4.2: The two branches of the diatomic chain. The **optical branch** is easier to excite with electromagnetic radiation; the **acoustic branch** is named after its (sound-like) linear dispersion at small k . There are as many branches as different atoms in the unit cell, but only one is acoustic (because of the constraint). The branches cannot cross because $\tilde{K}_{\alpha\beta}(k)$ is Hermitian (except in special cases).

The acoustic zero: $\sum_{\beta} \tilde{M}_{\alpha\beta}(0) = 0 \Rightarrow \sum_{\beta} \tilde{K}_{\alpha\beta}(0) \sqrt{m_{\beta}} = 0$, i.e. $\tilde{\phi}_{\beta} \propto \sqrt{m_{\beta}}$ is an eigenvector with $\omega = 0$. Finally

$$H = \sum_{k \in \mathcal{B}} \sum_{\lambda=1}^2 \hbar \omega_{\lambda}(k) \left(a_{k,\lambda}^{\dagger} a_{k,\lambda} + \frac{1}{2} \right),$$

$$q_{\alpha}(x) = \sum_{k \in \mathcal{B}} \sum_{\lambda=1}^2 \sqrt{\frac{\hbar}{2m_{\alpha} N \omega_{\lambda}(k)}} \left[\tilde{\phi}_{\alpha,\lambda}(k) a_{k,\lambda} e^{ikx} + \text{h.c.} \right],$$

and the same modifications apply to $p_{\alpha}(x)$.

4.2 Three-dimensional lattices

At low temperature the atoms form a periodic arrangement: a **motif** repeated on a lattice

$$\mathcal{L} = \{ \vec{b}_1 n_1 + \vec{b}_2 n_2 + \vec{b}_3 n_3 \mid n_{1,2,3} \in \mathbb{Z} \},$$

where the $\vec{b}_i$ are the **generators** or **primitive vectors** (linearly independent). The sublattices are $\mathcal{L}_i = \mathcal{L} + \vec{r}_i$. (Number of branches: a 1D diatomic lattice has 2; a 3D diatomic lattice has $3 \times 2 = 6$.)

From now on the Einstein summation convention is used:

$$H = \sum_{x \in \mathcal{L}} \frac{p_{\alpha}^i(x) \delta^{\alpha\beta} \delta_{ij} p_{\beta}^j(x)}{2m_{\alpha}} + \frac{1}{2} \sum_{x,y \in \mathcal{L}} q_{\alpha}^i(x) M_{ij}^{\alpha\beta}(x-y) q_{\beta}^j(y).$$

Periodic boundary conditions: $\vec{x} + \vec{b}_i N_i \equiv \vec{x}$ (no summation), with $N = N_1 N_2 N_3$ the total number of elements of the quotient lattice. The translations $T_i \phi(\vec{x}) = \phi(\vec{x} + \vec{b}_i)$ commute, $[T_i, T_j] = 0$, so they can be diagonalised simultaneously:

$$T_i \frac{e^{i\vec{k} \cdot \vec{x}}}{\sqrt{N}} = e^{i\vec{k} \cdot \vec{b}_i} \frac{e^{i\vec{k} \cdot \vec{x}}}{\sqrt{N}}, \quad e_{\vec{k}}(\vec{x}) = \frac{e^{i\vec{k} \cdot \vec{x}}}{\sqrt{N}}.$$

- Boundary conditions: $e_{\vec{k}}(\vec{x} + \vec{b}_i N_i) = e_{\vec{k}}(\vec{x}) \Rightarrow \vec{k} \cdot \vec{b}_i N_i \in 2\pi\mathbb{Z}$, so

$$\vec{k} \cdot \vec{b}_i \in \frac{2\pi}{N_i} \mathbb{Z}, \quad \mathcal{R} = \left\{ \vec{k} \mid \vec{k} \cdot \vec{b}_i \in \frac{2\pi}{N_i} \mathbb{Z} \right\}.$$

- Periodicity of reciprocal space: $e_{\vec{k}+\vec{g}}(\vec{x}) = e_{\vec{k}}(\vec{x}) \iff e^{i\vec{g}\cdot\vec{x}} = 1$ for all $\vec{x} \in \mathcal{L}$, so $\vec{g} \cdot \vec{b}_i \in 2\pi\mathbb{Z}$:

$$\mathcal{L}^* = \{ \vec{g} \mid \vec{g} \cdot \vec{b}_i \in 2\pi\mathbb{Z} \}$$

is the **reciprocal lattice**.

The primitive vectors of the reciprocal lattice satisfy $\vec{G}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$, and

$$\begin{aligned} \mathcal{L}^* &= \{ \vec{g} = n_1\vec{G}_1 + n_2\vec{G}_2 + n_3\vec{G}_3 \mid n_{1,2,3} \in \mathbb{Z} \}, \\ \mathcal{R} &= \left\{ \vec{k} = \frac{n_1}{N_1}\vec{G}_1 + \frac{n_2}{N_2}\vec{G}_2 + \frac{n_3}{N_3}\vec{G}_3 \right\}, \\ \mathcal{B} &= \mathcal{R}/\mathcal{L}^*, \end{aligned}$$

i.e. the Brillouin zone is obtained by identifying the elements of $\mathcal{R}$ connected by $\mathcal{L}^*$.

Wigner–Seitz cell. A convenient choice of $\mathcal{B}$ is the set of points closer to Γ ($\vec{k} = 0$) than to any other point of $\mathcal{L}^*$: it is bounded by the perpendicular bisectors of the segments joining Γ to its neighbours.

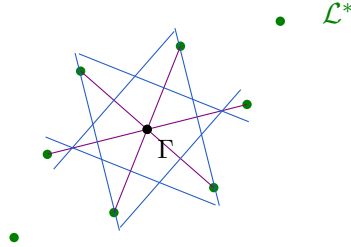
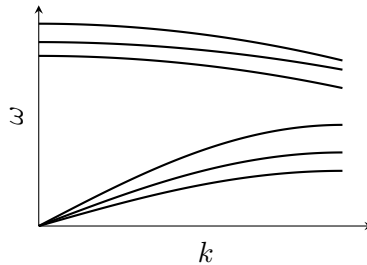


Figure 4.3: Construction of the Wigner–Seitz cell in a 2D reciprocal lattice.

Eigenvalue problem.

$$\begin{aligned} \sum_y \frac{M_{ij}^{\alpha\beta}(x-y)}{\sqrt{m_\alpha m_\beta}} \phi_\beta^j(y) &= \omega^2 \phi_\alpha^i(x), & \phi_\alpha^i(\vec{x}) &= e^{i\vec{k}\cdot(\vec{x}+\vec{r}_\alpha)} u_{\alpha,\lambda}^i(\vec{k}) \\ \implies K_{\alpha\beta}^{ij} u_{\beta,\lambda}^j &= \omega_\lambda^2 u_{\alpha,\lambda}^i, \end{aligned}$$

which for a diatomic 3D crystal is a 6×6 matrix problem: 3 acoustic and 3 optical branches, whose details depend on the restrictions (symmetries).



4.3 Polyatomic one-dimensional lattice

$$H = \sum_{x \in \mathcal{L}} \sum_{\alpha=1}^n \frac{p_\alpha^2(x)}{2m_\alpha} + \frac{1}{2} \sum_{x,y} \sum_{\alpha,\beta} q_\alpha(x) M_{\alpha\beta}(x-y) q_\beta(y);$$

the potential needs to be diagonalised twice.

First diagonalisation: (x, y) . With

$$q_\alpha(x) = \frac{1}{N} \sum_{k \in \mathcal{B}} e^{ikx} \tilde{q}_\alpha(k), \quad p_\alpha(x) = \frac{1}{N} \sum_{k \in \mathcal{B}} e^{ikx} \tilde{p}_\alpha(k),$$

$$\sum_y \sum_\beta M_{\alpha\beta}(x-y) q_\beta(y) = \frac{1}{N} \sum_k \sum_\beta \underbrace{\sum_y M_{\alpha\beta}(x-y) e^{ik(y-x)} e^{ikx}}_{\tilde{M}_{\alpha\beta}(k)} \tilde{q}_\beta(k),$$

and using $\sum_x e^{i(k+p)x} = N \delta_{k,-p}$,

$$H = \frac{1}{N} \sum_k \sum_\alpha \frac{\tilde{p}_\alpha(k) \tilde{p}_\alpha(-k)}{2m_\alpha} + \frac{1}{2N} \sum_k \sum_{\alpha,\beta} \tilde{q}_\alpha(-k) \sqrt{m_\alpha} \underbrace{\frac{\tilde{M}_{\alpha\beta}(k)}{\sqrt{m_\alpha m_\beta}}}_{\tilde{\mathcal{M}}_{\alpha\beta}(k)} \sqrt{m_\beta} \tilde{q}_\beta(k).$$

Second diagonalisation: (α, β) . Write

$$\tilde{q}_\alpha(k) = \sum_\lambda \frac{1}{\sqrt{m_\alpha}} u_{\alpha\lambda}(k) \tilde{\phi}_\lambda(k),$$

where u is chosen such that $u^\dagger \tilde{\mathcal{M}} u = \mathcal{K}$ is diagonal. Since $\tilde{q}_\alpha(-k) = \tilde{q}_\alpha^*(k)$ and $\tilde{\phi}_\lambda^*(k) = \tilde{\phi}_\lambda(-k)$, we get $u_{\alpha\lambda}^*(k) = u_{\alpha\lambda}(-k)$. For the transformation to be canonical,

$$\tilde{p}_\alpha(k) = \sum_\lambda \sqrt{m_\alpha} u_{\alpha\lambda}(k) \tilde{\psi}_\lambda(k),$$

with

$$[\tilde{q}_\alpha(k), \tilde{p}_\beta(p)] = i\hbar N \delta_{\alpha\beta} \delta_{k,-p}, \quad [\tilde{\phi}_\lambda(k), \tilde{\psi}_\mu(p)] = i\hbar N \delta_{\lambda\mu} \delta_{k,-p}.$$

Then, using unitarity $\sum_\alpha u_{\alpha\lambda_1}^* u_{\alpha\lambda_2} = \delta_{\lambda_1\lambda_2}$ and $u^\dagger \tilde{\mathcal{M}} u = \text{diag}(\mathcal{K}_\lambda)$,

$$H = \frac{1}{2N} \sum_k \sum_\lambda \tilde{\psi}_\lambda(k) \tilde{\psi}_\lambda(-k) + \frac{1}{2N} \sum_k \sum_\lambda \mathcal{K}_\lambda \tilde{\phi}_\lambda(k) \tilde{\phi}_\lambda(-k).$$

Balancing the two terms, $\frac{\hbar^2}{2\ell_\lambda^2} \sim \frac{\mathcal{K}_\lambda \ell_\lambda^2}{2}$, gives

$$\ell_\lambda^2 = \frac{\hbar}{\sqrt{\mathcal{K}_\lambda}}, \quad \hbar\omega_\lambda = \mathcal{K}_\lambda \ell_\lambda^2 \Rightarrow \omega_\lambda = \sqrt{\mathcal{K}_\lambda}.$$

With the dimensionless variables $\tilde{\psi}_\lambda(k) = \frac{\hbar}{\ell_\lambda} \sqrt{N} \pi_\lambda(k)$, $\tilde{\phi}_\lambda(k) = \ell_\lambda \sqrt{N} \varphi_\lambda(k)$,

$$H = \sum_{k \in \mathcal{B}} \sum_{\lambda=1}^n \frac{\hbar\omega_\lambda}{2} (\pi_\lambda(k) \pi_\lambda(-k) + \varphi_\lambda(k) \varphi_\lambda(-k)),$$

and with $a_{\lambda,k}^\dagger = \frac{1}{\sqrt{2}}(\varphi_\lambda(-k) - i\pi_\lambda(-k))$, $a_{\lambda,k} = \frac{1}{\sqrt{2}}(\varphi_\lambda(k) + i\pi_\lambda(k))$,

$$H = \sum_{k \in \mathcal{B}} \sum_{\lambda=1}^n \hbar\omega_\lambda \left(a_{\lambda,k}^\dagger a_{\lambda,k} + \frac{1}{2} \right).$$

For the displacements,

$$q_\alpha(x) = \frac{1}{N} \sum_k \tilde{q}_\alpha(k) e^{ikx} = \frac{1}{N} \sum_k \sum_\lambda \frac{u_{\alpha\lambda}(k)}{\sqrt{m_\alpha}} \ell_\lambda \sqrt{N} \varphi_\lambda(k) e^{ikx}$$

$$= \sum_k \sum_\lambda \sqrt{\frac{\hbar}{2Nm_\alpha\omega_\lambda(k)}} \left(u_{\alpha\lambda}(k) a_{\lambda,k} e^{ikx} + u_{\alpha\lambda}(k) a_{\lambda,-k}^\dagger e^{ikx} \right),$$

$$q_\alpha(x) = \sum_{k \in \mathcal{B}} \sum_{\lambda=1}^n \sqrt{\frac{\hbar}{2Nm_\alpha\omega_\lambda(k)}} \left(u_{\alpha\lambda}(k) a_{\lambda,k} e^{ikx} + u_{\alpha\lambda}^*(k) a_{\lambda,k}^\dagger e^{-ikx} \right),$$

and similarly for $p_\alpha(x)$.

4.4 Polyatomic three-dimensional lattice

With the Einstein summation convention,

$$H = \sum_{x \in \mathcal{L}} \frac{p_\alpha^i(x) \delta^{\alpha\beta} \delta_{ij} p_\beta^j(x)}{2\sqrt{m_\alpha m_\beta}} + \frac{1}{2} \sum_{x,y \in \mathcal{L}} q_\alpha^i(x) \sqrt{m_\alpha} \underbrace{\frac{M_{ij}^{\alpha\beta}(x-y)}{\sqrt{m_\alpha m_\beta}}}_{\mathcal{M}_{ij}^{\alpha\beta}(x-y)} \sqrt{m_\beta} q_\beta^j(y).$$

Diagonalisation in (x, y) . With

$$q_\alpha^i(\vec{x}) = \frac{1}{N} \sum_{\vec{k} \in \mathcal{B}} \frac{\tilde{q}_\alpha^i(\vec{k})}{\sqrt{m_\alpha}} e^{i\vec{k} \cdot (\vec{x} + \vec{r}_\alpha)}, \quad p_\alpha^i(\vec{x}) = \frac{1}{N} \sum_{\vec{k} \in \mathcal{B}} \sqrt{m_\alpha} \tilde{p}_\alpha^i(\vec{k}) e^{i\vec{k} \cdot (\vec{x} + \vec{r}_\alpha)},$$

$$\sum_{y \in \mathcal{L}} \mathcal{M}_{ij}^{\alpha\beta}(x-y) e^{i\vec{k} \cdot (\vec{y} + \vec{r}_\beta)} = \underbrace{\sum_{y \in \mathcal{L}} \mathcal{M}_{ij}^{\alpha\beta}(x-y) e^{i\vec{k} \cdot (\vec{y} + \vec{r}_\beta - \vec{x} - \vec{r}_\alpha)}}_{\tilde{\mathcal{M}}_{ij}^{\alpha\beta}(\vec{k})} e^{i\vec{k} \cdot (\vec{x} + \vec{r}_\alpha)},$$

one finds

$$H = \frac{1}{2N} \sum_{\vec{k}} \tilde{p}_\alpha^i(\vec{k}) \tilde{p}_\alpha^i(-\vec{k}) + \frac{1}{2N} \sum_{\vec{k}} \tilde{q}_\alpha^i(-\vec{k}) \tilde{\mathcal{M}}_{ij}^{\alpha\beta}(\vec{k}) \tilde{q}_\beta^j(\vec{k}).$$

Diagonalisation in $(\alpha i, \beta j)$. $\tilde{q}_\alpha^i(\vec{k}) = u_{\alpha,\lambda}^i(\vec{k}) \tilde{\phi}_\lambda(\vec{k})$ with $u^\dagger \tilde{\mathcal{M}} u = \text{diag}(\mathcal{K}_\lambda)$; the reality condition $\tilde{q}_\alpha^i(-\vec{k}) = \tilde{q}_\alpha^{i*}(\vec{k})$ gives $u_{\alpha,\lambda}^i(-\vec{k}) = u_{\alpha,\lambda}^{i*}(\vec{k})$, and for the transformation to be canonical $\tilde{p}_\alpha^i = u_{\alpha,\lambda}^i \tilde{\psi}_\lambda$. Then

$$H = \frac{1}{2N} \sum_{\vec{k} \in \mathcal{B}} \sum_{\lambda=1}^n \left(\tilde{\psi}_\lambda(\vec{k}) \tilde{\psi}_\lambda(-\vec{k}) + \mathcal{K}_\lambda \tilde{\phi}_\lambda(-\vec{k}) \tilde{\phi}_\lambda(\vec{k}) \right)$$

(here n is three times the number of atoms in the motif). Exactly as in 1D, $\ell_\lambda^2 = \hbar/\sqrt{\mathcal{K}_\lambda}$, $\omega_\lambda = \sqrt{\mathcal{K}_\lambda}$, $\tilde{\psi}_\lambda = \frac{\hbar}{\ell_\lambda} \sqrt{N} \pi_\lambda$, $\tilde{\phi}_\lambda = \ell_\lambda \sqrt{N} \varphi_\lambda$, and with

$$\pi_\lambda(\vec{k}) = \frac{i}{\sqrt{2}} (a_{-\vec{k},\lambda}^\dagger - a_{\vec{k},\lambda}), \quad \varphi_\lambda(\vec{k}) = \frac{1}{\sqrt{2}} (a_{-\vec{k},\lambda}^\dagger + a_{\vec{k},\lambda}),$$

$$H = \sum_{\vec{k} \in \mathcal{B}} \sum_{\lambda=1}^n \hbar \omega_\lambda(\vec{k}) \left(a_{\vec{k},\lambda}^\dagger a_{\vec{k},\lambda} + \frac{1}{2} \right),$$

$$q_\alpha^i(\vec{x}) = \sum_{\vec{k} \in \mathcal{B}} \sum_{\lambda=1}^n \sqrt{\frac{\hbar}{2N m_\alpha \omega_\lambda(\vec{k})}} \left[u_{\alpha,\lambda}^i(\vec{k}) a_{\vec{k},\lambda} e^{i\vec{k} \cdot (\vec{x} + \vec{r}_\alpha)} + u_{\alpha,\lambda}^{i*}(\vec{k}) a_{\vec{k},\lambda}^\dagger e^{-i\vec{k} \cdot (\vec{x} + \vec{r}_\alpha)} \right],$$

and similarly for $p_\alpha^i(\vec{x})$.

5 Processes: Fermi's golden rule and neutron scattering

Two ways of exploring the crystal: probing by scattering, and kinetic theory. The (quasi-)momentum $\vec{k}$ is conserved when the system is translation invariant, and in the harmonic approximation the occupation numbers $n_{\vec{k}_1}, \dots, n_{\vec{k}_N}$ are, in principle, conserved quantities. A perturbation changes them, $n_{\vec{k}_i} \rightarrow n_{\vec{k}_i} \pm 1$.

5.1 Fermi's golden rule

Let $H = H_0 + V$ with V small, $H_0 |\lambda\rangle = E_\lambda |\lambda\rangle$, $\langle \lambda | \lambda \rangle = 1$. Start in $|i\rangle = |\lambda\rangle$ and evolve, $|f\rangle = \mathcal{U}(t) |i\rangle$. The **transition probability** is

$$w_{\lambda \rightarrow \lambda'} := |\langle \lambda' | f \rangle|^2 = \left| \langle \lambda' | \underbrace{e^{iH_0 t/\hbar} \mathcal{U}(t)}_{\tilde{\mathcal{U}}(t)} | \lambda \rangle \right|^2, \quad \tilde{\mathcal{U}}(t) = \mathcal{T} e^{-\frac{i}{\hbar} \int_0^t dt' \tilde{V}(t')} \simeq \mathbb{1} - \frac{i}{\hbar} \int_0^t dt' \tilde{V}(t') + \dots$$

Assuming $\langle \lambda' | \lambda \rangle = 0$,

$$w_{\lambda \rightarrow \lambda'} = \frac{1}{\hbar^2} \left| \langle \lambda' | \int_0^t dt' e^{iE_{\lambda'} t'/\hbar} V e^{-iE_\lambda t'/\hbar} | \lambda \rangle \right|^2 = \frac{1}{\hbar^2} |\langle \lambda' | V | \lambda \rangle|^2 \left| \int_0^t dt' e^{i\Delta E t'/\hbar} \right|^2,$$

with $\Delta E = E_{\lambda'} - E_\lambda$ and $\int_0^t dt' e^{i\Delta E t'/\hbar} = \frac{\hbar}{i\Delta E} (e^{i\Delta E t/\hbar} - 1)$, of modulus $\frac{2\hbar}{\Delta E} \sin \frac{\Delta E t}{2\hbar}$. Hence

$$w_{\lambda \rightarrow \lambda'} = |\langle \lambda' | V | \lambda \rangle|^2 \frac{4}{(\Delta E)^2} \sin^2 \left(\frac{\Delta E t}{2\hbar} \right).$$

It is more interesting (and useful for experiments) to integrate over a window of final states, $\int w_{\lambda \rightarrow \lambda'} d\mu_{\lambda'}$, which leads to integrals like

$$\int_{-\infty}^{\infty} dz \frac{\sin^2(zt)}{z^2} f(z)$$

with f a smooth function. For $z \gg 1/t$ the integrand oscillates rapidly, while $\int dz f(z)/z^2$ would diverge at small z : as $t \rightarrow \infty$ the kernel concentrates at $z = 0$ and

$$\int_{-\infty}^{\infty} dz \frac{\sin^2(zt)}{z^2} f(z) \simeq f(0) \pi t, \quad \text{i.e.} \quad \frac{\sin^2(zt)}{z^2} \rightarrow \pi t \delta(z).$$

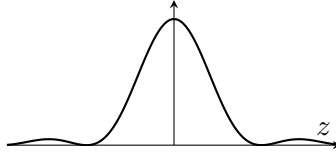


Figure 5.1: The kernel $\sin^2(zt)/z^2$ (here for $t = 1$): height t^2 , width $\sim 1/t$, area πt .

Then

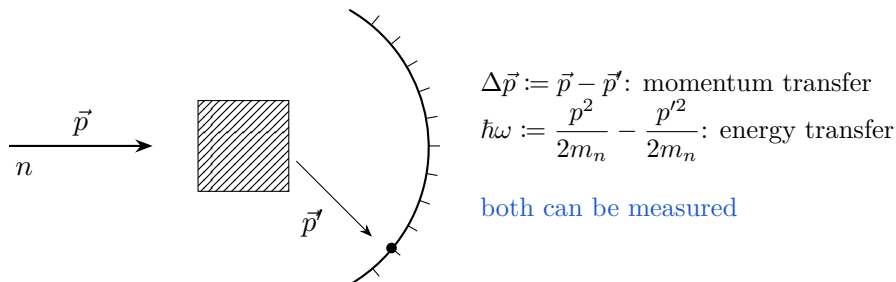
$$\frac{4}{(\Delta E)^2} \sin^2 \left(\frac{\Delta E t}{2\hbar} \right) \simeq 4 \delta(\Delta E) \frac{t\pi}{2\hbar} = \frac{2\pi t}{\hbar} \delta(\Delta E),$$

and the **transition rate** is **Fermi's golden rule**:

$$\Gamma_{\lambda \rightarrow \lambda'} := \frac{w_{\lambda \rightarrow \lambda'}}{t} = \frac{2\pi}{\hbar} |\langle \lambda' | V | \lambda \rangle|^2 \delta(E_{\lambda'} - E_\lambda).$$

It only makes sense when integrated over final states.

5.2 Application: neutron scattering experiment



The number of neutrons detected at given $\Delta\vec{p}$ and energy transfer gives the **scattering data**: what is the physics inside the box?

- $\Delta E = 0$: elastic scattering;
- $\Delta E \neq 0$: inelastic scattering.

Thermal neutrons have wavelengths $\lambda \sim 0.1$ nm, comparable to the lattice spacing, so they explore the Brillouin zone of the material; their energy $\varepsilon \sim 0.025$ eV is small, so they do not disturb the structure of the system much (they do not interact much with the material).

The model. The interaction between the neutron (at $\vec{r}$) and a single atom (at $\vec{r}_A$) is $V = C\delta(\vec{r} - \vec{r}_A)$, where C is a constant that depends on the details of the interaction. For the lattice,

$$V = C \sum_A \delta(\vec{r} - \vec{r}_A) = C \sum_{\vec{x} \in \mathcal{L}_N} \delta(\vec{r} - \vec{x} - \vec{q}(\vec{x})).$$

The neutron crosses a sample of size L in $\Delta t \sim L/v_n$ (there might be a deflection), so $w \sim \Gamma_{\vec{p} \rightarrow \vec{p}'} \Delta t$. In what follows momenta in exponents are written as wave vectors. The initial and final states are

$$|i\rangle = |\lambda\rangle \otimes \frac{e^{i\vec{p}' \cdot \vec{r}}}{\sqrt{L^3}}, \quad |f\rangle = |\lambda'\rangle \otimes \frac{e^{i\vec{p} \cdot \vec{r}}}{\sqrt{L^3}},$$

where $|\lambda\rangle$ is a vibrational eigenstate of the lattice and $\vec{r}$, $\vec{q}$ are operators. Then

$$\Gamma_{\lambda\vec{p} \rightarrow \lambda'\vec{p}'} = \frac{2\pi C^2}{\hbar} \left| \frac{1}{L^3} \int d^3r e^{-i\vec{p}' \cdot \vec{r}} \langle \lambda' | \frac{V}{C} | \lambda \rangle e^{i\vec{p} \cdot \vec{r}} \right|^2 \delta(E_{\lambda'} + \varepsilon(\vec{p}') - E_\lambda - \varepsilon(\vec{p})),$$

and the integral over $\vec{r}$ gives

$$\langle \lambda' | \sum_{\vec{x} \in \mathcal{L}_N} e^{i\Delta\vec{p} \cdot \vec{x}} e^{i\Delta\vec{p} \cdot \vec{q}(\vec{x})} | \lambda \rangle,$$

where $\vec{q}(\vec{x})$ is the dynamical variable of the phonons. Therefore

$$\Gamma_{\lambda\vec{p} \rightarrow \lambda'\vec{p}'} = \frac{2\pi C^2}{\hbar L^6} \left| \sum_{\vec{x} \in \mathcal{L}_N} e^{i\Delta\vec{p} \cdot \vec{x}} \langle \lambda' | e^{i\Delta\vec{p} \cdot \vec{q}(\vec{x})} | \lambda \rangle \right|^2 \delta(\hbar\omega - (E_{\lambda'} - E_\lambda)).$$

By itself this is “theoretical garbage”: there is no way to prepare or measure the state of the crystal ($\sim 10^{23}$ particles). One averages over the initial states with the thermal weights $w(\lambda)$ and sums over the final ones:

$$\Gamma_{\vec{p} \rightarrow \vec{p}'} := \sum_{\lambda, \lambda'} w(\lambda) \Gamma_{\lambda\vec{p} \rightarrow \lambda'\vec{p}'} = \frac{2\pi C^2}{\hbar L^6} S(\Delta\vec{p}, \omega),$$

$$S(\Delta\vec{p}, \omega) = \sum_{\lambda, \lambda'} w(\lambda) \left| \sum_{\vec{x} \in \mathcal{L}_N} e^{i\Delta\vec{p} \cdot \vec{x}} \langle \lambda' | e^{i\Delta\vec{p} \cdot \vec{q}(\vec{x})} | \lambda \rangle \right|^2 \delta(\hbar\omega - (E_{\lambda'} - E_\lambda)),$$

the **dynamic structure factor**.

5.3 Inelastic scattering

At $T = 0$ the initial state is the ground state. We are interested in creating one phonon, $|\lambda'\rangle = a_{\vec{Q},s}^\dagger |0\rangle$, so we need $\langle 0 | a_{\vec{Q},s} e^{i\Delta\vec{p} \cdot \vec{q}(\vec{x})} | 0 \rangle$, with (monoatomic lattice, mass m)

$$\vec{q}(\vec{x}) = \sum_{\vec{Q}} \sum_s \sqrt{\frac{\hbar}{2Nm\omega_s(\vec{Q})}} [\vec{u}_{\vec{Q},s} a_{\vec{Q},s} e^{i\vec{Q} \cdot \vec{x}} + \text{h.c.}] = \vec{q}_- + \vec{q}_+,$$

where $\vec{q}_-$ ($\vec{q}_+$) contains the annihilation (creation) operators, sum included. Since $[\vec{q}_-, \vec{q}_+]$ is a number, $e^{A+B} = e^A e^B e^{-\frac{1}{2}[A,B]}$ gives

$$e^{i\Delta\vec{p}\cdot\vec{q}} = e^{i\Delta\vec{p}\cdot\vec{q}_+} e^{i\Delta\vec{p}\cdot\vec{q}_-} e^{-W}, \quad W(\Delta\vec{p}) = \frac{1}{2}[\Delta\vec{p}\cdot\vec{q}_-, \Delta\vec{p}\cdot\vec{q}_+] = \frac{1}{2} \sum_{\vec{Q},s} \frac{\hbar |\vec{u}_{\vec{Q},s} \cdot \Delta\vec{p}|^2}{2mN\omega_s(\vec{Q})}.$$

The factor $e^{i\Delta\vec{p}\cdot\vec{q}_-}$ acts trivially on the vacuum, and using $[a, f(a^\dagger)] = df/da^\dagger$,

$$\langle 0 | a_{\vec{Q},s} e^{i\Delta\vec{p}\cdot\vec{q}_+} | 0 \rangle = \langle 0 | [a_{\vec{Q},s}, e^{i\Delta\vec{p}\cdot\vec{q}_+}] | 0 \rangle = i \sqrt{\frac{\hbar}{2Nm\omega_s(\vec{Q})}} (\vec{u}_{\vec{Q},s}^* \cdot \Delta\vec{p}) e^{-i\vec{Q}\cdot\vec{x}}.$$

Hence

$$S = \sum_{\vec{Q},s} \frac{\hbar |\vec{u}_{\vec{Q},s} \cdot \Delta\vec{p}|^2}{2m\omega_s(\vec{Q})N} \underbrace{\left| \sum_{\vec{x} \in \mathcal{L}_N} e^{i(\Delta\vec{p}-\vec{Q})\cdot\vec{x}} \right|^2}_{N^2 \sum_{\vec{g} \in \mathcal{L}^*} \delta_{\Delta\vec{p}-\vec{Q},\vec{g}}} \delta(\omega - \omega_s(\vec{Q})) e^{-2W},$$

$$S = \sum_s \frac{N\hbar |\vec{u}_{\Delta\vec{p},s} \cdot \Delta\vec{p}|^2 e^{-2W(\Delta\vec{p})}}{2m\omega_s(\Delta\vec{p})} \delta(\omega - \omega_s(\Delta\vec{p})),$$

with $\Delta\vec{p}$ understood modulo the reciprocal lattice. Measuring the peaks of S as a function of $\Delta\vec{p}$ and ω maps out the dispersion relations.

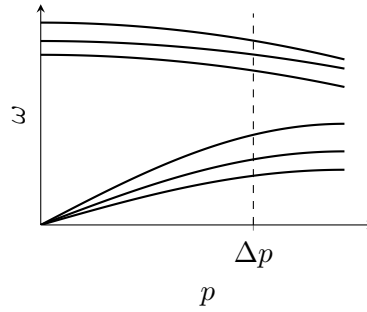


Figure 5.2: Dispersion relations probed by inelastic neutron scattering.

Elastic case.

$$S = \left| \sum_{\vec{x} \in \mathcal{L}_N} e^{i\Delta\vec{p}\cdot\vec{x}} \right|^2 e^{-2W},$$

where e^{-2W} is the **Debye–Waller factor**. It can be used to “see” the zero-point motion. At finite temperature,

$$W = \sum_{s,\vec{Q}} \frac{\hbar |\vec{u}_{\vec{Q},s} \cdot \Delta\vec{p}|^2}{4mN\omega_s(\vec{Q})} \coth\left(\frac{\hbar\omega_s(\vec{Q})}{2T}\right),$$

which reduces to the $T = 0$ expression above as $T \rightarrow 0$.

6 Superfluidity

6.1 Superfluid helium: experiments

Can we achieve a liquid state at zero temperature? We need a light element and weak interactions, in order to destroy the lattice structure. Hydrogen is light but its interactions are not weak

enough; helium satisfies both: it is a good candidate.

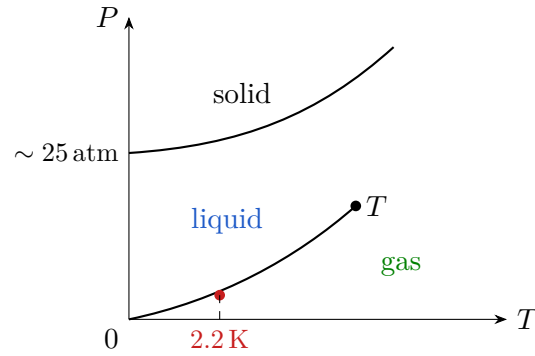
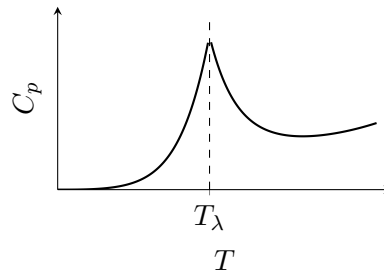


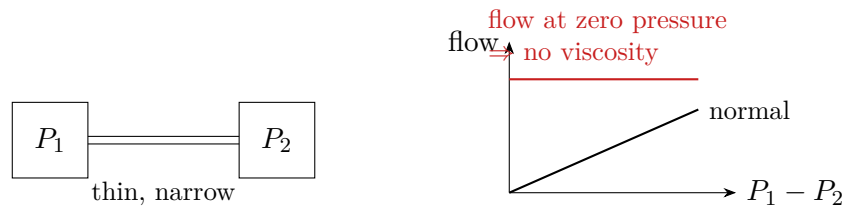
Figure 6.1: Phase diagram of ^4He . The easiest way to cool it down is by moving along the coexistence line TO (the system is always boiling).

In 1937 Kapitza, and Allen (with Misener), observed that at 2.2 K the boiling suddenly stops.

Thermodynamic properties (Kapitza). The specific heat C_p has a “ λ ”-shaped peak at $T_\lambda = 2.2$ K. (Lower temperature means less available phase space.)



Capillary channel. Two vessels at pressures P_1 , P_2 connected by a thin, narrow channel: below T_λ there is flow at zero pressure difference (no viscosity), whereas a normal fluid has a flow proportional to $P_1 - P_2$.



Rotating annulus. Liquid set into rotation in a ring-shaped vessel: after a long time, the velocity of the fluid remains the same. No viscosity $\Rightarrow$ **superfluidity**.

Torsion pendulum. A disc oscillating inside the liquid is damped: there *is* viscosity (the system is stationary).

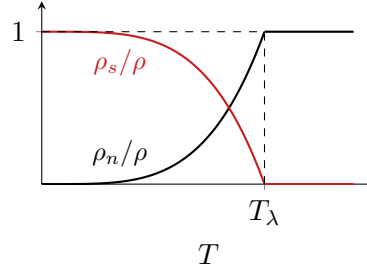
Rotating bucket. For a rigidly rotating normal fluid $\vec{v} = \vec{\Omega} \times \vec{r}$, so $\nabla \times \vec{v} = 2\vec{\Omega}$; the superfluid instead has $\nabla \times \vec{v} = 0$ and rotates by forming vortices, whose number is proportional to the angular momentum.

Two-fluid model (Tisza, 1938).

$$\rho = \rho_s + \rho_n, \quad \chi(T, p) := \frac{\rho_n}{\rho} :$$

a mixture of two different fluids, superfluid and normal.

Andronikashvili experiment. A stack of closely spaced discs performs torsional oscillations in the liquid; only the normal component is dragged along, so the frequency $\omega = \sqrt{\kappa/I}$ (with I the moment of inertia) measures χ .



6.2 The Landau criterion (1941)

The ground state is superfluid. Viscosity means that momentum of the walls is transferred to the fluid, by creating an elementary excitation of momentum $\vec{p}$ and energy $\varepsilon(\vec{p})$. For a fluid of total mass M moving with velocity $\vec{v} = \vec{P}/M$ relative to the walls,

$$\vec{P} = \vec{P}' + \vec{p}, \quad \frac{P^2}{2M} = \frac{P'^2}{2M} + \varepsilon(\vec{p}) \implies \left(\frac{\vec{P} + \vec{P}'}{2M} \right) \cdot \vec{p} = \varepsilon(\vec{p}),$$

and for $M \rightarrow \infty$ we get the **excitation condition**

$$\vec{v} \cdot \vec{p} = \varepsilon(\vec{p}).$$

Excitations can only be created if $v \geq v_c = \min_p \varepsilon(p)/p$: below this critical velocity the flow is dissipationless.

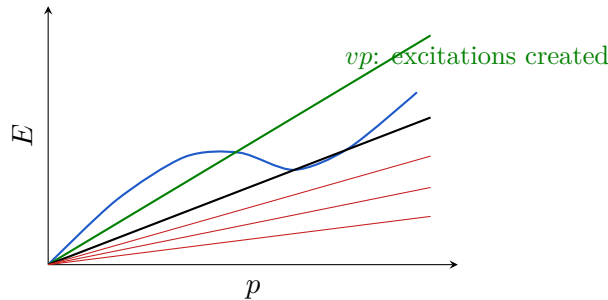


Figure 6.2: Excitation spectrum (blue) and lines $E = vp$. For small v (red lines) there is no intersection: no excitations can be created.

^3He vs ^4He . ^3He , being lighter, seems to be a better candidate, but it becomes superfluid only at $\sim \text{mK}$ (a different kind of superfluidity). Why does ^3He not show the behaviour of ^4He ? ^3He has spin $\frac{1}{2}$ (a fermion), ^4He has spin 0 (a boson).

6.3 The ideal Bose gas

Bosons can all occupy the single-particle ground state; an elementary excitation promotes one of them. In a box of side L with periodic boundary conditions,

$$H = \sum_{\vec{k} \in \frac{2\pi}{L}\mathbb{Z}^3} n_{\vec{k}} \frac{\hbar^2 k^2}{2m}, \quad N = \sum_{\vec{k}} n_{\vec{k}}; \quad \text{ground state: } n_0 = N.$$

Since $\varepsilon(k) = \hbar^2 k^2 / 2m$, $v_c = \min \varepsilon / p = 0$: the ideal gas ground state is not a good model of a superfluid.

$$Z_N = \sum_{\lambda=\{n_\alpha\}} e^{-\beta \sum_\alpha \varepsilon_\alpha n_\alpha} = \sum_\lambda \prod_\alpha e^{-\beta \varepsilon_\alpha n_\alpha} = \prod_\alpha \sum_{n_\alpha=0}^{\infty} e^{-\varepsilon_\alpha n_\alpha / T} \Big|_{\sum_\alpha n_\alpha = N}.$$

The constraint is removed by the **grand partition function** (Laplace transform of Z_N):

$$\Xi(\mu) = \sum_{N=0}^{\infty} e^{\mu N / T} Z_N = \prod_\alpha \sum_{n_\alpha=0}^{\infty} e^{-(\varepsilon_\alpha - \mu) n_\alpha / T} = \prod_\alpha \frac{1}{1 - e^{-\beta(\varepsilon_\alpha - \mu)}}.$$

The **Landau potential** (grand potential) of this non-interacting system is a sum over modes,

$$\Omega(\mu, T, V) = -T \ln \Xi(\mu) = \sum_\alpha \Omega_\alpha, \quad d\Omega = -P dV - S dT - N d\mu,$$

$$-\frac{\partial \Omega}{\partial \mu} = \sum_\alpha \frac{1}{e^{\beta(\varepsilon_\alpha - \mu)} - 1} = \sum_\alpha n_\alpha = N, \quad n_\alpha := \frac{1}{e^{(\varepsilon_\alpha - \mu)/T} - 1}$$

— the **Bose–Einstein distribution**. Thus

$$N = \sum_{\vec{k} \in \frac{2\pi}{L}\mathbb{Z}^3} \frac{1}{e^{(\varepsilon_k - \mu)/T} - 1} \quad (\mu < 0).$$

At $T = 0$: $n_0 = N$, $n_{\vec{k}} = 0$ for $|\vec{k}| > 0$ (ground state).

For $T > 0$,

$$n_0 = \frac{1}{e^{-\mu/T} - 1} = \frac{T}{|\mu|} + \dots \implies |\mu| \simeq \frac{T}{N} \sim \frac{T}{L^3},$$

which is much smaller than the level spacing $\hbar^2/(2mL^2)$. So μ is close to zero for the excited states, and

$$\underbrace{N - N_0}_{\text{number of excitations}} = \sum_{k \neq 0} \frac{1}{e^{\varepsilon_k/T} - 1} = V \int d\varepsilon \rho(\varepsilon) \frac{1}{e^{\varepsilon/T} - 1}, \quad \rho(\varepsilon) \sim \sqrt{\varepsilon}$$

(similar to phonons in a crystal), so

$$N - N_0 = VC T^{3/2} \int dz \frac{\sqrt{z}}{e^z - 1} = VC' T^{3/2}.$$

The condensate is empty, $N_0 = 0$, at the critical temperature

$$N = VC' T_c^{3/2};$$

above this temperature the condensate disappears.

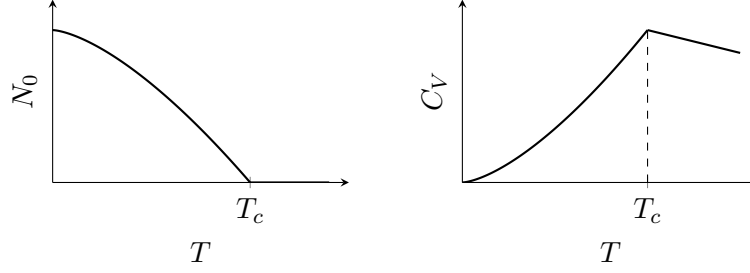


Figure 6.3: Condensate occupation and specific heat of the ideal Bose gas.

Next step: incorporate interactions.

7 Second quantisation

First quantisation, one particle. The particle lives in a **configuration space** $\mathcal{C}$; for example, for a particle with spin S , $\mathcal{C} = \mathbb{R}^3 \times \{-S, -S+1, \dots, S-1, S\}$. A wave function is a map $\psi : \mathcal{C} \rightarrow \mathbb{C}$, and the Hilbert space is $\mathcal{V} = \{\psi(x) \mid x \in \mathcal{C}\}$: if $\psi_1, \psi_2 \in \mathcal{V}$ then $a\psi_1 + b\psi_2 \in \mathcal{V}$, with inner product

$$(\psi_1, \psi_2) := \sum_{x \in \mathcal{C}} \psi_1^*(x) \psi_2(x).$$

Two identical particles. $\psi : \mathcal{C} \times \mathcal{C} \rightarrow \mathbb{C}$, with $X = \{x_1, x_2\}$ and $\mathcal{V}^2 = \{\psi(X) \mid X \in \mathcal{C} \times \mathcal{C}\} = \mathcal{V} \otimes \mathcal{V}$ (Cartesian product of configuration spaces).

Permutations. The permutation of configuration space $\pi\{x_1, x_2\} := \{x_2, x_1\}$ defines the permutation operator $P\psi(X) := \psi(\pi X)$. What does “identical” mean? The same wave function up to a phase:

$$\text{identical} \iff P\psi = \lambda\psi.$$

Since $P^2 = \mathbb{I}$, $\lambda = 1$ (**bosons**, integer S) or $\lambda = -1$ (**fermions**, half-integer S). There is no clear explanation of the connection with spin within this framework: it is the spin–statistics theorem of QFT.

N identical particles. $\mathcal{V}^N = \{\psi(X) \mid X \in \mathcal{C} \times \dots \times \mathcal{C}\} = \bigotimes^N \mathcal{V}$. Let π_{ij} be the elementary permutation of the i -th and j -th variables and $P_{ij}\psi(X) := \psi(\pi_{ij}X)$ the elementary permutation operator. There exist two simultaneous eigenspaces:

$$\forall i, j : P_{ij}\psi = +\psi \text{ (bosons)}, \quad \forall i, j : P_{ij}\psi = -\psi \text{ (fermions)}$$

— a much more powerful statement than the one given by Fermi (the exclusion principle).

7.1 Independent bosons

The Hamiltonian of independent bosons is a so-called **one-body operator**, $H = \sum_{i=1}^N H_i$, where

$$H_i\psi(x_1, \dots, x_i, \dots, x_N) = \sum_{y_i} h(x_i, y_i)\psi(x_1, \dots, y_i, \dots, x_N);$$

in general $H_i = P_{1i}H_1P_{1i}$. Assume the single-particle spectrum

$$\sum_y h(x, y)\chi_\alpha(y) = E_\alpha\chi_\alpha(x), \quad \alpha \in \mathcal{B}$$

(here $\mathcal{B}$ is the set of quantum numbers). The product state

$$\Psi_A(X) := \chi_{\alpha_1}(x_1)\chi_{\alpha_2}(x_2)\cdots\chi_{\alpha_N}(x_N), \quad X = \{x_1, \dots, x_N\}, \quad A = \{\alpha_1, \dots, \alpha_N\},$$

is formally an eigenstate of the Hamiltonian, $H\Psi_A = E_A\Psi_A$ with $E_A = E_{\alpha_1} + \cdots + E_{\alpha_N}$, but it does not obey the right symmetry: $P\Psi_A \neq \Psi_A$.

Symmetrisation. Since $[P, H] = 0$ for any permutation operator ($P\Psi(X) = \Psi(\pi X)$), $H(P\Psi_A) = E_A(P\Psi_A)$. The totally symmetric state has the same energy and the bosonic symmetry:

$$\Phi_A(X) = \mathcal{S}\Psi_A(X) := \frac{1}{\sqrt{N!}} \sum_{P \in S_N} P\Psi_A(X).$$

Complications:

- The totally symmetric function contains $N!$ terms: intractable for large systems.
- Perturbation-theory calculations require integrals of products, i.e. $N! \times N!$ integrals.

7.2 Second quantisation

For any given single-particle Hamiltonian, the states Φ_A form a complete basis of the symmetric many-particle space. In this basis we define the **occupation number operator**

$$\hat{n}_\alpha \Phi_A := n_\alpha \Phi_A,$$

with n_α the number of times α is encountered in A . The Hamiltonian of independent bosons is then

$$H = \sum_{\alpha \in \mathcal{B}} E_\alpha \hat{n}_\alpha,$$

similar to the usual harmonic-oscillator energy.

Creation and annihilation operators. For each $\alpha \in \mathcal{B}$ one defines a representation of the occupation number operator in the familiar form

$$\hat{n}_\alpha := a_\alpha^\dagger a_\alpha, \quad [a_\alpha, a_\beta^\dagger] = \delta_{\alpha\beta},$$

with the ket notation $|\lambda\rangle := \Phi_A(X)$, $\lambda = \{n_\alpha \mid \alpha \in \mathcal{B}\}$ — as for phonons. Most operators can be represented as a polynomial sum of a and $a^\dagger$.

Rules of second quantisation.

- One-body operators:

$$\hat{V}^{(1)} = \sum_{i=1}^N V_i \quad \longrightarrow \quad \hat{V}^{(1)} = \sum_{\alpha, \beta \in \mathcal{B}} V_{\beta\alpha} a_\beta^\dagger a_\alpha, \quad V_{\beta\alpha} := \sum_{x_1 \in \mathcal{C}} \chi_\beta^*(x_1) V_1 \chi_\alpha(x_1).$$

(If an operator O does not commute with the permutations, $O\Phi_A(X)$ is not a bosonic wave function.)

- Two-body operators, $\hat{V}_{12}\psi(x_1, x_2, \dots, x_N) = \sum_{y_1, y_2} v_{12}(x_1, x_2, y_1, y_2)\psi(y_1, y_2, x_3, \dots, x_N)$, with $V_{ji} = V_{ij}$ (V_{ii} would be a one-body operator):

$$\hat{V}^{(2)} = \sum_{i < j} \hat{V}_{ij} \quad \longrightarrow \quad \hat{V}^{(2)} = \frac{1}{2} \sum_{\alpha, \beta, \gamma, \delta \in \mathcal{B}} V_{\beta\delta, \alpha\gamma} a_\beta^\dagger a_\delta^\dagger a_\alpha a_\gamma,$$

$$V_{\beta\delta, \alpha\gamma} := \sum_{x_1, x_2 \in \mathcal{C}} \chi_\beta^*(x_1) \chi_\delta^*(x_2) \hat{V}_{12} \chi_\alpha(x_1) \chi_\gamma(x_2),$$

which automatically symmetrises V .

7.3 Quantum field

A **quantum field** is an operator-valued function on the configuration space, $\phi : \mathcal{C} \rightarrow \text{End}(\mathcal{V} \otimes \mathcal{V} \otimes \dots \otimes \mathcal{V})$:

$$\phi(x) := \sum_{\alpha \in \mathcal{B}} \chi_{\alpha}(x) a_{\alpha}.$$

Properties:

- From the completeness of the basis of orbitals,

$$[\phi(x), \phi^{\dagger}(x')] = \sum_{\alpha} \chi_{\alpha}(x) \chi_{\alpha}^{*}(x') = \delta(x, x')$$

(a delta function in the continuous part of $\mathcal{C}$, a Kronecker symbol in the discrete part).

- From orthonormality, $\sum_{x \in \mathcal{C}} \phi^{\dagger}(x) \phi(x) = \sum_{\alpha \in \mathcal{B}} a_{\alpha}^{\dagger} a_{\alpha} = N$.

Using the quantum field operator,

$$\begin{aligned} V^{(1)} = \sum_{i=1}^N V_i &\longrightarrow V^{(1)} = \sum_{x_1 \in \mathcal{C}} \phi^{\dagger}(x_1) \hat{V}_1 \phi(x_1), \\ V^{(2)} = \sum_{i < j} V_{ij} &\longrightarrow V^{(2)} = \frac{1}{2} \sum_{x_1, x_2 \in \mathcal{C}} \phi^{\dagger}(x_1) \phi^{\dagger}(x_2) \hat{V}_{12} \phi(x_2) \phi(x_1). \end{aligned}$$

Example: total momentum. $P = \sum_{j=1}^N (-i \nabla_j)$ ($\hbar = 1$):

$$\begin{aligned} P &= \int d^3x \phi^{\dagger}(x) (-i \nabla) \phi(x) = \int d^3x \sum_{\alpha} \chi_{\alpha}^{*}(x) a_{\alpha}^{\dagger} (-i \nabla) \sum_{\beta} \chi_{\beta}(x) a_{\beta} \\ &= \sum_{\alpha, \beta \in \mathcal{B}} a_{\alpha}^{\dagger} a_{\beta} \int d^3x \chi_{\alpha}^{*}(x) (-i \nabla) \chi_{\beta}(x). \end{aligned}$$

8 Bogoliubov's theory of superfluidity

A good example of an emergent phenomenon.

8.1 Interacting bosons

$$H = \sum_{i=1}^N \frac{p_i^2}{2m} + \sum_{i < j} U(\vec{r}_i - \vec{r}_j)$$

in a box of side L , volume $\mathcal{V} = L^3$. Single-particle basis:

$$\chi_{\vec{k}}(\vec{r}) = \frac{e^{i\vec{k} \cdot \vec{r}}}{\sqrt{\mathcal{V}}}, \quad \vec{k} \in \frac{2\pi}{L} \mathbb{Z}^3$$

(periodic boundary conditions: not necessary, but useful). The second-quantised Hamiltonian is

$$H = \sum_{\vec{k}} \frac{\hbar^2 k^2}{2m} a_{\vec{k}}^{\dagger} a_{\vec{k}} + \frac{1}{2} \sum_{\vec{k}_1 \vec{k}_2 \vec{k}_3 \vec{k}_4} U_{\vec{k}_1 \vec{k}_2, \vec{k}_3 \vec{k}_4} a_{\vec{k}_1}^{\dagger} a_{\vec{k}_2}^{\dagger} a_{\vec{k}_4} a_{\vec{k}_3},$$

where

$$\begin{aligned}
U_{\vec{k}_1 \vec{k}_2, \vec{k}_3 \vec{k}_4} &= \int d\vec{r}_1 \int d\vec{r}_2 \frac{1}{\mathcal{V}^2} e^{i(\vec{k}_3 - \vec{k}_1) \cdot \vec{r}_1 + i(\vec{k}_4 - \vec{k}_2) \cdot \vec{r}_2} U(\vec{r}_1 - \vec{r}_2) \\
&= \frac{1}{\mathcal{V}} \frac{1}{\mathcal{V}} \underbrace{\int d\vec{r}_2 e^{i(\vec{k}_3 + \vec{k}_4 - \vec{k}_1 - \vec{k}_2) \cdot \vec{r}_2}}_{\delta_{\vec{k}_1 + \vec{k}_2, \vec{k}_3 + \vec{k}_4}} \underbrace{\int d(\vec{r}_1 - \vec{r}_2) e^{-i(\vec{k}_1 - \vec{k}_3) \cdot (\vec{r}_1 - \vec{r}_2)} U(\vec{r}_1 - \vec{r}_2)}_{\tilde{U}(\vec{k}_1 - \vec{k}_3)}, \\
U_{\vec{k}_1 \vec{k}_2, \vec{k}_3 \vec{k}_4} &= \frac{1}{\mathcal{V}} \delta_{\vec{k}_1 + \vec{k}_2, \vec{k}_3 + \vec{k}_4} \tilde{U}(\vec{k}_1 - \vec{k}_3) :
\end{aligned}$$

momentum conservation, because the interaction term is translation invariant. Hence

$$H = \sum_{\vec{k}} \frac{\hbar^2 k^2}{2m} a_{\vec{k}}^\dagger a_{\vec{k}} + \frac{1}{2\mathcal{V}} \sum_{\vec{k}_1, \vec{k}_2, \vec{q}} \tilde{U}(\vec{q}) a_{\vec{k}_1 + \vec{q}}^\dagger a_{\vec{k}_2 - \vec{q}}^\dagger a_{\vec{k}_2} a_{\vec{k}_1}.$$

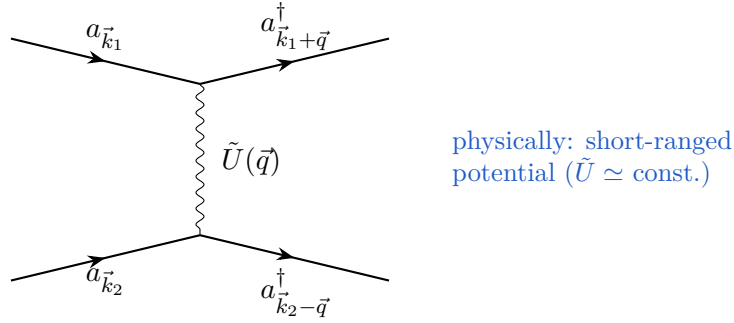


Figure 8.1: The interaction vertex.

8.2 Semiclassical condensate

For a short-ranged potential, $\tilde{U}(\vec{q}) \simeq U_0$:

$$H = \sum_{\vec{k}} \frac{\hbar^2 k^2}{2m} a_{\vec{k}}^\dagger a_{\vec{k}} + \frac{U_0}{2\mathcal{V}} \sum_{\vec{k}_1, \vec{k}_2, \vec{q}} a_{\vec{k}_1 + \vec{q}}^\dagger a_{\vec{k}_2 - \vec{q}}^\dagger a_{\vec{k}_2} a_{\vec{k}_1}$$

(solved analytically only in 1D: Lieb–Liniger). The unperturbed system is a condensate; assume that weak perturbations do not destroy it. Then

$$a_0^\dagger a_0 \simeq N_0 \gg 1, \quad a_0 \simeq \sqrt{N_0} + b_0, \quad a_0^\dagger \simeq \sqrt{N_0} + b_0^\dagger,$$

and the interaction terms can be ordered by powers of N_0 :

- all four legs in the condensate: $\sim N_0^2$;
- exactly one leg with $\vec{k} \neq 0$: not allowed (momentum conservation);
- two legs in the condensate, $a_{\vec{k}}^\dagger a_{\vec{k}}$ with $\tilde{U}(0)$ or $\tilde{U}(\vec{k})$: $\sim N_0$;
- two condensate legs on the same side: $\sim N_0 a_{\vec{k}}^\dagger a_{-\vec{k}}^\dagger$ (and its conjugate).

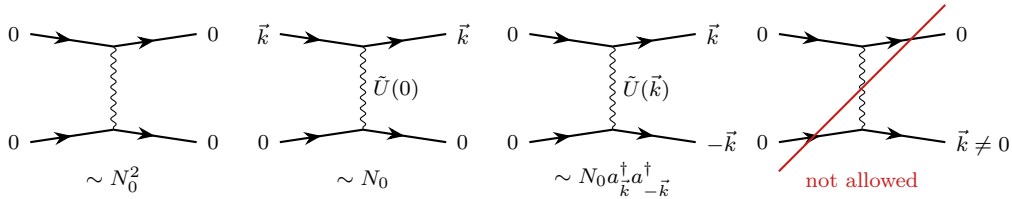


Figure 8.2: The first (trivial) Feynman diagrams of the expansion around the condensate.

8.3 Bogoliubov's mean-field Hamiltonian

We expand in the large parameter N_0 . Writing $\varepsilon_k = \hbar^2 k^2 / 2m$, $\rho_0 = N_0 / \mathcal{V}$, and

$$H = \varepsilon_0 a_0^\dagger a_0 + \sum_{\vec{k} \neq 0} \varepsilon_k a_{\vec{k}}^\dagger a_{\vec{k}} + \frac{U_0}{2\mathcal{V}} \left[\sum_{\vec{k}_1, \vec{k}_2} a_{\vec{k}_1}^\dagger a_{\vec{k}_2}^\dagger a_{\vec{k}_2} a_{\vec{k}_1} + \sum_{\vec{q} \neq 0} a_{\vec{k}_1 + \vec{q}}^\dagger a_{\vec{k}_2 - \vec{q}}^\dagger a_{\vec{k}_2} a_{\vec{k}_1} \right],$$

we keep the terms with at least two condensate operators:

$$H = \sum_{\vec{k}} \varepsilon_k a_{\vec{k}}^\dagger a_{\vec{k}} + \frac{U_0}{2\mathcal{V}} \left[(a_0^\dagger)^2 (a_0)^2 + \sum_{\vec{q} \neq 0} (a_{\vec{q}}^\dagger a_{-\vec{q}}^\dagger a_0^2 + (a_0^\dagger)^2 a_{-\vec{q}} a_{\vec{q}}) + \sum_{\vec{k} \neq 0} (4 a_{\vec{k}}^\dagger a_0^\dagger a_0 a_{\vec{k}}) + \dots \right],$$

where the factor 4 collects the direct ($\tilde{U}(0)$) and exchange ($\tilde{U}(\vec{k})$) terms, each appearing twice. With $(a_0^\dagger)^2 a_0^2 \simeq N_0^2$ and $a_0 \simeq \sqrt{N_0}$,

$$H_{\text{MF}} \simeq \frac{U_0 N_0^2}{2\mathcal{V}} + \sum_{\vec{k} \neq 0} (\varepsilon_k + 2\rho_0 U_0) a_{\vec{k}}^\dagger a_{\vec{k}} + \frac{\rho_0 U_0}{2} \sum_{\vec{k} \neq 0} (a_{-\vec{k}} a_{\vec{k}} + a_{\vec{k}}^\dagger a_{-\vec{k}}^\dagger).$$

Using $N_0 = N - \sum_{\vec{k} \neq 0} a_{\vec{k}}^\dagger a_{\vec{k}}$, i.e. $N_0^2 \simeq N^2 - 2N \sum_{\vec{k} \neq 0} a_{\vec{k}}^\dagger a_{\vec{k}}$ and $\rho_0 \simeq \rho = N/\mathcal{V}$ at this order,

$$H_{\text{MF}} = \frac{N^2 U_0}{2\mathcal{V}} + \sum'_{\vec{k} \neq 0} \left[(\varepsilon_k + \rho U_0) (a_{\vec{k}}^\dagger a_{\vec{k}} + a_{-\vec{k}}^\dagger a_{-\vec{k}}) + \rho U_0 (a_{-\vec{k}} a_{\vec{k}} + a_{\vec{k}}^\dagger a_{-\vec{k}}^\dagger) \right],$$

where $\sum'$ is taken over half of momentum space (e.g. $k_z > 0$).

Change of variables. $H = \sum'_{\vec{k}} H_{\vec{k}}$ with

$$H_{\vec{k}} = \varepsilon_0 (a_{\vec{k}}^\dagger a_{\vec{k}} + a_{-\vec{k}}^\dagger a_{-\vec{k}}) + \varepsilon_1 (a_{\vec{k}} a_{-\vec{k}} + a_{-\vec{k}}^\dagger a_{\vec{k}}^\dagger), \quad \varepsilon_0 = \varepsilon_k + \rho U_0, \quad \varepsilon_1 = \rho U_0$$

(here ε_0 no longer denotes the energy of the $\vec{k} = 0$ state): two coupled harmonic oscillators. They are solved by the **Bogoliubov transformation**

$$a_{\vec{k}} = u b_{\vec{k}} + v b_{-\vec{k}}^\dagger, \quad a_{-\vec{k}} = u b_{-\vec{k}} + v b_{\vec{k}}^\dagger,$$

which mixes creation and annihilation operators with the same sense of momentum (a natural construction). The transformation needs to be canonical:

$$[b_{\vec{k}}, b_{\vec{k}}^\dagger] = 1, \quad [b_{-\vec{k}}, b_{-\vec{k}}^\dagger] = 1, \quad [b_{\vec{k}}, b_{-\vec{k}}] = [b_{\vec{k}}^\dagger, b_{-\vec{k}}^\dagger] = 0 \quad (\Rightarrow u^2 - v^2 = 1),$$

and one also wants the Hamiltonian to take the form $H_{\vec{k}} = \tilde{\varepsilon} (b_{\vec{k}}^\dagger b_{\vec{k}} + b_{-\vec{k}}^\dagger b_{-\vec{k}})$ (up to a constant). This gives

$$u^2 = \frac{1}{2} \left(\frac{\varepsilon_0}{\tilde{\varepsilon}} + 1 \right), \quad v^2 = \frac{1}{2} \left(\frac{\varepsilon_0}{\tilde{\varepsilon}} - 1 \right), \quad \tilde{\varepsilon} = \sqrt{\varepsilon_0^2 - \varepsilon_1^2} = \sqrt{\varepsilon_k^2 + 2\rho U_0 \varepsilon_k},$$

$$H_{\vec{k}} = \tilde{\varepsilon}_k (b_{\vec{k}}^\dagger b_{\vec{k}} + b_{-\vec{k}}^\dagger b_{-\vec{k}}) + \text{const.}$$

The momentum is $P = \sum_{\vec{k}} \hbar \vec{k} a_{\vec{k}}^\dagger a_{\vec{k}} = \sum_{\vec{k}} \hbar \vec{k} b_{\vec{k}}^\dagger b_{\vec{k}}$.

Definition of the vacuum. $|\text{vac}\rangle$: $b_{\vec{k}}|\text{vac}\rangle = 0$ for all $\vec{k}$. First excited state: $|\vec{k}\rangle = b_{\vec{k}}^\dagger|\text{vac}\rangle$.

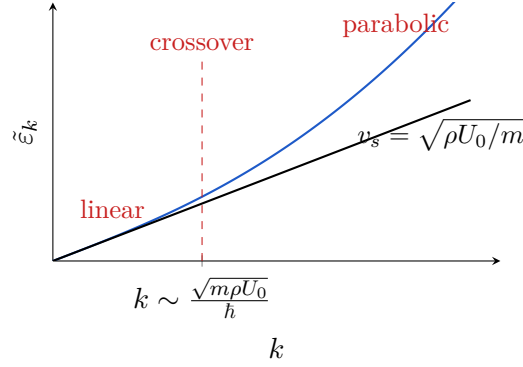


Figure 8.3: Quasiparticle dispersion $\tilde{\epsilon}_k = \sqrt{\epsilon_k^2 + 2\rho U_0 \epsilon_k}$, $\epsilon_k = \hbar^2 k^2 / 2m$: linear (sound) at small k , parabolic at large k . By the Landau criterion, the system is a superfluid.

8.4 Condensate depletion

The basis of Bogoliubov's particles is not the same as that of the constituent particles: the Bogoliubov vacuum has a finite density of constituent particles with finite momentum,

$$n_{\vec{k}} = \langle \text{vac} | a_{\vec{k}}^\dagger a_{\vec{k}} | \text{vac} \rangle = v_k^2 = \frac{1}{2} \left(\frac{\epsilon_k + \rho U_0}{\sqrt{\epsilon_k^2 + 2\rho U_0 \epsilon_k}} - 1 \right).$$

The density of particles not in the condensate is

$$\Delta\rho = \frac{N - N_0}{\mathcal{V}} = \frac{1}{\mathcal{V}} \sum_{\vec{k} \neq 0} n_{\vec{k}} = \frac{1}{3\pi^2} \left(\frac{mv_s}{\hbar} \right)^3.$$

The theory breaks down when this quantity becomes comparable with the density of particles. Therefore the validity of the theory is controlled by the parameter

$$\gamma = \frac{1}{\rho} \left(\frac{mv_s}{\hbar} \right)^3.$$

9 The Gross–Pitaevskii equation

Summary of Bogoliubov's theory: it

- constructs the ground state of the condensate;
- constructs the elementary excitations;
- demonstrates superfluidity through the Landau criterion;
- does *not* describe the hydrodynamics (macroscopic inhomogeneous motion of large masses of fluid).

Approach: identify collective variables and extend the analysis.

9.1 Semiclassical theory

Example, the harmonic oscillator: $a, a^\dagger, N = a^\dagger a, H = \hbar\omega a^\dagger a$. For states with large occupation numbers,

$$a \simeq \sqrt{N} e^{i\theta}, \quad a^\dagger \simeq \sqrt{N} e^{-i\theta} \quad \implies \quad |[a, a^\dagger]| \ll \|aa^\dagger\|,$$

and the equations of motion

$$\dot{a} = -i\omega a, \quad \dot{a}^* = i\omega a^* \iff \dot{N} = 0, \quad \dot{\theta} = -\omega.$$

In the condensate, $n_{\vec{k}} = v_{\vec{k}}^2 \gg 1$ for $k < \sqrt{\rho U_0 m}/\hbar$: the long-wavelength modes can be treated as classical variables. Define

$$\psi(\vec{x}) = \sum_{\vec{k}} \frac{e^{i\vec{k}\cdot\vec{x}}}{\sqrt{\mathcal{V}}} a_{\vec{k}}, \quad \psi^\dagger(\vec{x}) = \sum_{\vec{k}} \frac{e^{-i\vec{k}\cdot\vec{x}}}{\sqrt{\mathcal{V}}} a_{\vec{k}}^\dagger, \quad [\psi(\vec{x}), \psi^\dagger(\vec{y})] = \delta(\vec{x} - \vec{y});$$

for the long-wavelength modes, $\psi(x)$ and $\psi^*(x)$ are classical.

The Hamiltonian.

$$H = \sum_{i=1}^N \left(-\frac{\hbar^2}{2m} \Delta_i \right) + U_0 \sum_{i < j} \delta(\vec{x}_i - \vec{x}_j) \implies H = \int d^3x \left[\psi^\dagger \left(-\frac{\hbar^2}{2m} \Delta \right) \psi + \frac{U_0}{2} \psi^\dagger \psi^\dagger \psi \psi \right].$$

It is convenient to subtract a constant (conserved) term, $H \rightarrow H - \mu N$; this does not change the dynamics, but the ground state contains a finite number of particles. In the classical limit the Hamiltonian becomes the energy

$$H = \int d^3x \left[\frac{\hbar^2}{2m} |\nabla \psi|^2 + \frac{U_0}{2} |\psi|^4 - \mu |\psi|^2 \right].$$

9.2 The equations of motion

From $H = \int d^3x \left[\psi^\dagger \left(-\frac{\hbar^2}{2m} \Delta \right) \psi + \frac{U_0}{2} \psi^\dagger \psi^\dagger \psi \psi - \mu \psi^\dagger \psi \right]$ and $[\psi(\vec{x}), \psi^\dagger(\vec{y})] = \delta(\vec{x} - \vec{y})$,

$$i\hbar \partial_t \psi = -[H, \psi] = -\frac{\hbar^2}{2m} \Delta \psi + (U_0 \psi^\dagger \psi - \mu) \psi,$$

the **quantum non-linear Schrödinger equation** (with an exact solution in 1D). Neglecting the non-commutativity we obtain the **Gross–Pitaevskii equation**:

$$\begin{aligned} i\hbar \partial_t \psi &= -\frac{\hbar^2}{2m} \Delta \psi + (U_0 |\psi|^2 - \mu) \psi, \\ -i\hbar \partial_t \psi^* &= -\frac{\hbar^2}{2m} \Delta \psi^* + (U_0 |\psi|^2 - \mu) \psi^*. \end{aligned}$$

The GP action.

$$S = \int dt \int d^3x \left[\underbrace{\psi^* i\hbar \partial_t \psi}_{\text{symplectic term}} - \mathcal{U}(\psi, \psi^*) \right], \quad \mathcal{U}(\psi, \psi^*) = \frac{\hbar^2}{2m} |\nabla \psi|^2 + \frac{U_0}{2} |\psi|^4 - \mu |\psi|^2.$$

$\delta S = 0$ gives the GP equations; the action is nicer for discussing symmetries.

Symmetries and conserved quantities.

- translations, $\vec{x} \rightarrow \vec{x} + \delta\vec{x}$, $t \rightarrow t + \delta t$: momentum and energy;
- rotations, $\vec{x} \rightarrow (1 + \epsilon)\vec{x}$: angular momentum;
- global $U(1)$, $\psi \rightarrow e^{i\theta}\psi$, $\psi^* \rightarrow e^{-i\theta}\psi^*$: the $U(1)$ current,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0, \quad \rho := \psi^* \psi, \quad \vec{j} := \frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*).$$

Notice that ρ is just the density of particles ($\int \rho = N$).

The equilibrium solution. The ground state of $i\hbar\partial_t\psi = -\frac{\hbar^2}{2m}\Delta\psi + (U_0|\psi|^2 - \mu)\psi$ is

$$\psi = \sqrt{\frac{\mu}{U_0}} e^{i\theta},$$

a **broken-symmetry state**: it minimises $\mathcal{V}(\psi, \psi^*) = \frac{U_0}{2}|\psi|^4 - \mu|\psi|^2$, which has a whole circle of minima.

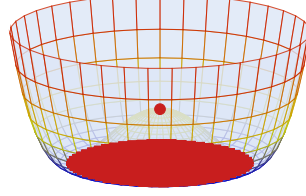


Figure 9.1: The “Mexican hat” potential $\mathcal{V}(\psi, \psi^*)$ over the plane $(\text{Re } \psi, \text{Im } \psi)$: the symmetric point $\psi = 0$ is a maximum; the ground states lie on the circle $|\psi|^2 = \mu/U_0$.

9.3 Hard and soft modes

Semiclassically, the condensate is a broken-symmetry state of $U(1)$: motion in the ρ direction is hard, motion in the θ direction is “easy”. Change variables:

$$\psi(\vec{x}, t) = \sqrt{\rho + \hbar^{-1}w(\vec{x}, t)} e^{-i\theta(\vec{x}, t)} \implies S = \int dt \int d^3x [w\dot{\theta} - H(\theta, w)],$$

so w plays the role of the canonical variable associated with θ . For small w , in a gradient expansion, the canonical system has

$$H = \frac{\rho\hbar^2}{2m}(\nabla\theta)^2 + \frac{U_0}{2\hbar^2}w^2 + \text{h.o.t.}$$

The Legendre transformation, $\dot{\theta} = \partial H / \partial w = U_0 w / \hbar^2$, gives the **linear σ -model**

$$S = \hbar^2 \int dt \int d^3x \left[\frac{\dot{\theta}^2}{2U_0} - \frac{\rho}{2m}(\nabla\theta)^2 \right],$$

whose solutions are free waves with speed $v = \sqrt{\rho U_0 / m}$ (the Bogoliubov sound velocity).

9.4 Vortices

θ is not a usual field: $\theta \equiv \theta + 2\pi n$, $n \in \mathbb{Z}$. A static solution satisfies $\Delta\theta = 0$ (the wave equation with $\dot{\theta} = 0$). A solution is

$$\theta = n\phi, \quad n \in \mathbb{Z},$$

with ϕ the azimuthal angle around the z -axis: a **vortex**, with a singularity on the z -axis (resolved by $\psi = 0$ there). With $\vec{r}_\perp$ the distance vector from the axis,

$$\vec{j} = \frac{\hbar\rho}{m}\nabla\theta = \frac{\hbar\rho n}{m} \frac{\hat{z} \times \vec{r}_\perp}{r_\perp^2}, \quad \vec{v} = \frac{\hbar}{m}\nabla\theta = \frac{\hbar n}{m} \frac{\hat{z} \times \vec{r}_\perp}{r_\perp^2} \quad (\vec{j} = \rho\vec{v}),$$

$$\oint \vec{v} \cdot d\vec{l} = \frac{2\pi\hbar n}{m},$$

a **topological charge**: it does not depend on the path. The vortex is a **topological defect**, and this leads to the appearance of vortices in rotating superfluids.

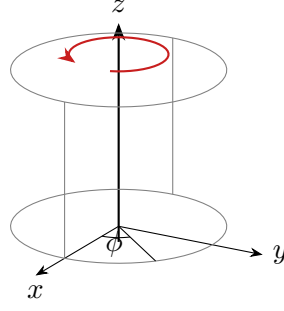


Figure 9.2: A vortex line along the z -axis.

10 Fermions

Fermionic fluids. Physical systems: ^3He ; mobile electrons in solids; ultracold atomic gases (^{40}K , ^6Li , ...); neutron stars.

As before, one particle has configuration space $\mathcal{C} = \mathbb{R}^3 \times \{-S, \dots, S\}$ and wave functions $\psi : \mathcal{C} \rightarrow \mathbb{C}$; two identical particles have $X = \{x_1, x_2\}$, $\mathcal{V}^2 = \{\psi(X) \mid X \in \mathcal{C} \times \mathcal{C}\} = \mathcal{V} \otimes \mathcal{V}$, and N identical particles $\mathcal{V}^N = \{\psi(X) \mid X \in \mathcal{C} \times \dots \times \mathcal{C}\} = \mathcal{V} \otimes \dots \otimes \mathcal{V}$ (N times).

10.1 Independent fermions

$H = H_1 + H_2 + \dots + H_N$. Using the basis $H_1 \chi_\alpha(x_1) = E_\alpha \chi_\alpha(x_1)$, $\alpha \in \mathcal{B}$ (the set of quantum numbers), the product state $\Psi_A(X) = \chi_{\alpha_1}(x_1) \dots \chi_{\alpha_N}(x_N)$ does not have the correct symmetry. The antisymmetrised state is the **Slater determinant**

$$\Phi_A(X) = \mathcal{A}\{\Psi_A(X)\} = \frac{1}{\sqrt{N!}} \det(S), \quad S_{ij} := \chi_{\alpha_i}(x_j), \quad A = \{\alpha_1, \dots, \alpha_N\}.$$

Implications:

- if $\alpha_i = \alpha_j$ for $i \neq j$, then $\Phi_A(X) = 0$, since S has two proportional rows: the **Pauli principle**;
- the wave function vanishes when two coordinates coincide.

10.2 Second quantisation

$\hat{n}_\alpha \Phi_A := n_\alpha \Phi_A$, with eigenvalues $n_\alpha \in \{0, 1\}$ (the number of times α is encountered in A), and

$$H = \sum_{\alpha \in \mathcal{B}} E_\alpha \hat{n}_\alpha.$$

Fermi oscillator and the Clifford algebra. With the anticommutator $\{A, B\} := AB + BA$, the algebra is formed by a pair of Hermitian-conjugate operators obeying

$$\{c, c^\dagger\} = 1, \quad \{c, c\} = \{c^\dagger, c^\dagger\} = 0$$

(c and $c^\dagger$ are nilpotent). The number operator $N = c^\dagger c$ satisfies $[N, c] = -c$, $[N, c^\dagger] = c^\dagger$. (It is a graded algebra: N is even, $c, c^\dagger$ are odd.)

The Hilbert space of a fermionic oscillator. $c|0\rangle := 0$ defines the vacuum, $N|0\rangle = 0$; $c^\dagger|0\rangle = |1\rangle$ is the only excited state, $N|1\rangle = |1\rangle$.

Independent fermionic oscillators: extended Clifford algebra.

$$\{c_\alpha, c_\beta\} = 0, \quad \{c_\alpha^\dagger, c_\beta^\dagger\} = 0, \quad \{c_\alpha, c_\beta^\dagger\} = \delta_{\alpha\beta}, \quad \alpha, \beta \in \mathcal{B},$$

with occupation number operators $N_\alpha := c_\alpha^\dagger c_\alpha$, the mathematical vacuum $c_\alpha |\text{vac}\rangle = 0$ for all $\alpha \in \mathcal{B}$, and $|\lambda\rangle = c_{\alpha_1}^\dagger c_{\alpha_2}^\dagger \cdots |\text{vac}\rangle$, $\lambda = \{N_\alpha \mid \alpha \in \mathcal{B}\}$.

Rules of second quantisation. With the field operator $\psi(x) := \sum_{\alpha \in \mathcal{B}} \chi_\alpha(x) c_\alpha$,

$$\begin{aligned} \text{one-body: } V^{(1)} &= \sum_{i=1}^N V_i = \sum_{x_1 \in \mathcal{C}} \psi^\dagger(x_1) V_1 \psi(x_1), \\ \text{two-body: } V^{(2)} &= \sum_{i < j} V_{ij} = \frac{1}{2} \sum_{x_1, x_2 \in \mathcal{C}} \psi^\dagger(x_1) \psi^\dagger(x_2) V_{12} \psi(x_2) \psi(x_1). \end{aligned}$$

Example: particle density operator. $\rho(x) = \sum_{i=1}^N \delta(x, x_i)$ becomes

$$\rho(x) = \int dx_1 \psi^\dagger(x_1) \delta(x - x_1) \psi(x_1) = \psi^\dagger(x) \psi(x) = \sum_{\alpha, \beta \in \mathcal{B}} c_\alpha^\dagger c_\beta \chi_\alpha^*(x) \chi_\beta(x)$$

in normal modes (orbitals).

10.3 Ground state and elementary excitations

For $H = \sum_{\alpha \in \mathcal{B}} E_\alpha n_\alpha$, the lowest-energy state is obtained by populating the single-particle states in order of increasing energy: the **Fermi sea**. The most useful definition is

$$c_\alpha |\text{FS}\rangle = 0 \quad (E_\alpha > E_F), \quad c_\alpha^\dagger |\text{FS}\rangle = 0 \quad (E_\alpha < E_F), \quad n_\alpha = \begin{cases} 1, & E_\alpha < E_F, \\ 0, & E_\alpha > E_F. \end{cases}$$

Elementary excitations (particle-hole pairs): $c_\alpha^\dagger c_\beta |\text{FS}\rangle$ with $E_\alpha > E_F$, $E_\beta < E_F$, of energy $E = E_\alpha - E_\beta$.

10.4 Thermodynamics

The grand partition function is

$$\Xi := \sum_{\lambda} e^{-(E_\lambda - \mu N)/T} = \prod_{\alpha \in \mathcal{B}} \sum_{n_\alpha=0}^1 e^{-(E_\alpha - \mu) n_\alpha / T} = \prod_{\alpha \in \mathcal{B}} (1 + e^{-(E_\alpha - \mu)/T}),$$

and the grand (Landau) potential

$$\Omega := -T \ln \Xi = \sum_{\alpha \in \mathcal{B}} \Omega_\alpha, \quad \Omega_\alpha = -T \ln(1 + e^{-(E_\alpha - \mu)/T}).$$

The **Fermi-Dirac distribution**:

$$n_\alpha = - \left(\frac{\partial \Omega_\alpha}{\partial \mu} \right)_T = \frac{1}{1 + e^{(E_\alpha - \mu)/T}},$$

with $n_\alpha \in (0, 1)$; μ has no restrictions. Some useful identities:

$$M = - \left(\frac{\partial \Omega}{\partial B} \right)_{\mu, T}, \quad \chi = \frac{\partial M}{\partial B} \Big|_{B=0}, \quad \Omega = -PV, \quad c_V = - \frac{T}{V} \frac{\partial^2 \Omega}{\partial T^2}.$$

10.5 Ideal Fermi gas

A collection of independent fermions with single-particle Hamiltonian $H = -\hbar^2 \nabla^2 / 2m$. Single-particle eigenstates:

$$\chi_{\vec{k},\sigma}(\vec{x}, s) = \frac{1}{\sqrt{\mathcal{V}}} e^{i\vec{k} \cdot \vec{x}} \delta_{\sigma,s}$$

(in general, spinors $\psi_{\uparrow}(s)$, $\psi_{\downarrow}(s)$). In a square box (just for convenience), $\vec{k} = \frac{2\pi}{L}(n, m, l)$, $n, m, l \in \mathbb{Z}$, $L^3 = \mathcal{V}$.

The ground state. The eigenstates of the many-body Hamiltonian have

$$E = \sum_{\vec{k},\sigma} \frac{\hbar^2 k^2}{2m} n_{\vec{k},\sigma}, \quad \text{subject to} \quad N = \sum_{\vec{k},\sigma} n_{\vec{k},\sigma};$$

the ground state fills the **Fermi sphere**. Its radius follows from (spin $\frac{1}{2}$)

$$\rho = \frac{1}{\mathcal{V}} \sum_{\vec{k},\sigma} n_{\vec{k},\sigma} = 2 \int_0^{k_F} \frac{4\pi k^2}{(2\pi)^3} dk = \frac{k_F^3}{3\pi^2}, \quad \varepsilon_F = \frac{\hbar^2 k_F^2}{2m}$$

(the Fermi energy). The Fermi sea can then be used as the vacuum state.

Thermodynamics.

$$\Omega = -T \sum_{\sigma} \sum_{\vec{k}} \ln(1 + e^{-(\varepsilon(\vec{k}) - \mu)/T}) = -\mathcal{V} T \int_{-\infty}^{\infty} d\varepsilon \rho(\varepsilon) \ln(1 + e^{-(\varepsilon - \mu)/T}),$$

with the density of states $\rho(\varepsilon) = \frac{1}{\mathcal{V}} (2S + 1) \sum_{\vec{k}} \delta(\varepsilon - \varepsilon(\vec{k}))$.

Specific heat.

$$c_V = -\frac{T}{V} \frac{\partial^2 \Omega}{\partial T^2} = \int_0^{\infty} d\varepsilon \rho(\varepsilon) \frac{(\varepsilon - \mu)^2}{4T^2} \frac{1}{\cosh^2(\frac{\varepsilon - \mu}{2T})}.$$

At low temperature the chemical potential is $\mu \simeq \varepsilon_F$ and only states near the Fermi energy contribute; with $\nu_F = \rho(\varepsilon_F)$ the density of states close to the Fermi energy,

$$c_V = \nu_F \int_{-\infty}^{\infty} d\varepsilon \frac{\varepsilon^2}{4T^2} \frac{1}{\cosh^2(\varepsilon/2T)} = \frac{\pi^2 T}{3} \nu_F.$$

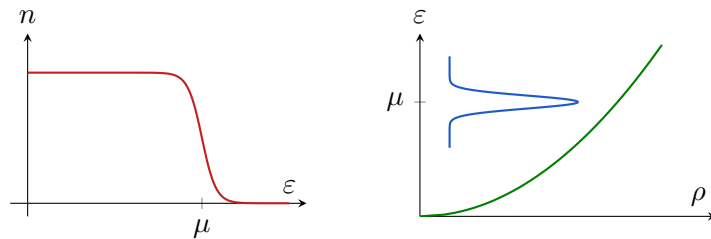


Figure 10.1: Left: Fermi-Dirac distribution at low temperature. Right: density of states $\rho(\varepsilon) \propto \sqrt{\varepsilon}$ and the window of width $\sim T$ around μ that contributes to c_V .

Magnetic susceptibility. The Zeeman term

$$H_Z = -\frac{\mu_B g}{\hbar} \vec{B} \cdot \vec{S}, \quad \mu_B := \frac{e\hbar}{2m_e c} \text{ (Bohr magneton),}$$

with g depending on the particle one is looking at, gives spin-dependent energy levels

$$\varepsilon_{\pm}(\vec{k}) = \varepsilon(\vec{k}) \pm \beta B, \quad \beta = \frac{\mu_B g}{2}.$$

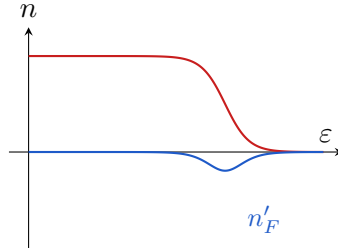
At low temperature,

$$\begin{aligned} \Omega &= -T \int d\varepsilon \rho(\varepsilon) \frac{\ln(1 + e^{-(\varepsilon + \beta B - \mu)/T}) + \ln(1 + e^{-(\varepsilon - \beta B - \mu)/T})}{2}, \\ \chi &:= \left. \frac{\partial M}{\partial B} \right|_{B=0} = \beta^2 \int d\varepsilon \rho(\varepsilon) \left(-\frac{\partial n_F}{\partial \varepsilon} \right) = \beta^2 \nu_F \int d\varepsilon \left(-\frac{dn_F}{d\varepsilon} \right) = \beta^2 \nu_F, \\ \chi &= \beta^2 \nu_F. \end{aligned}$$

With $c_V = \frac{\pi^2 T}{3} \nu_F$,

$$\frac{\chi}{c_V} = \frac{3}{T} \left(\frac{\beta}{\pi} \right)^2 :$$

the combination $T\chi/c_V$ is a universal constant, independent of the system (the **Wilson ratio**).



(Further reading on the GP equation: *Bose–Einstein Condensation in Dilute Gases*, Pethick & Smith.)

11 Weakly interacting Fermi gas

Motivation: explore qualitatively the effects of two-body interactions, and describe qualitatively physical systems such as ultracold gases.

Weak interactions. Crudely, interactions are weak if in every two-body collision process the interaction potential is much smaller than the kinetic energy. In a Fermi gas this translates into

$$V_{\text{int}} \ll E_F \sim \rho^{2/3}.$$

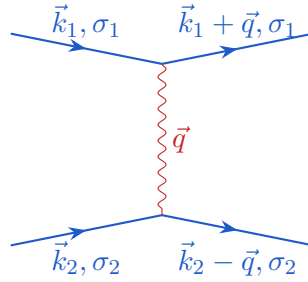
For the Coulomb interaction $V_{\text{int}} \sim e^2 \rho^{1/3}$, so in a Fermi gas the high-density limit normally corresponds to the limit of weak interactions. The relevant parameter is r_s , the mean inter-particle distance in units of the Bohr radius; in metals $r_s \sim 2$ –5.

11.1 The Hamiltonian

$$H = \int d^3x \sum_{\sigma} \psi_{\sigma}^{\dagger}(x) \left[-\frac{\hbar^2}{2m} \nabla^2 - \mu \right] \psi_{\sigma}(x) + \frac{1}{2} \int d^3x_1 \int d^3x_2 \sum_{\sigma_1, \sigma_2} \psi_{\sigma_1}^{\dagger}(x_1) \psi_{\sigma_2}^{\dagger}(x_2) U(x_1 - x_2) \psi_{\sigma_2}(x_2) \psi_{\sigma_1}(x_1).$$

It is translationally invariant; U does not depend on the spin (there is no spin exchange at a vertex). In the plane-wave basis,

$$H = \sum_{\vec{k}, \sigma} \xi_k c_{\vec{k}, \sigma}^{\dagger} c_{\vec{k}, \sigma} + \frac{1}{2\mathcal{V}} \sum_{\vec{k}_1, \vec{k}_2, \vec{q}} \sum_{\sigma_1, \sigma_2} \tilde{U}(\vec{q}) c_{\vec{k}_1 + \vec{q}, \sigma_1}^{\dagger} c_{\vec{k}_2 - \vec{q}, \sigma_2}^{\dagger} c_{\vec{k}_2, \sigma_2} c_{\vec{k}_1, \sigma_1}, \quad \xi_k := \frac{\hbar^2 k^2}{2m} - \mu.$$



Energy levels in perturbation theory. $H = H_0 + V$, $E_n = E_n^{(0)} + \delta E_n$; in leading-order (non-degenerate) perturbation theory, $\delta E_n = \langle E_n | V | E_n \rangle$.

11.2 Hartree–Fock theory

Use Slater-determinant states $|\lambda\rangle = c_{\vec{k}_1, \sigma_1}^{\dagger} \cdots c_{\vec{k}_N, \sigma_N}^{\dagger} |0\rangle$.

The Wick theorem. General statement:

$$\langle \lambda | c_{\alpha_1}^{\dagger} \cdots c_{\alpha_N}^{\dagger} c_{\beta_N} \cdots c_{\beta_1} | \lambda \rangle = \det[\langle c_{\alpha_i}^{\dagger} c_{\beta_j} \rangle].$$

In particular,

$$\langle c_a^{\dagger} c_b^{\dagger} c_e c_f \rangle = \langle c_b^{\dagger} c_e \rangle \langle c_a^{\dagger} c_f \rangle - \langle c_a^{\dagger} c_e \rangle \langle c_b^{\dagger} c_f \rangle :$$

each *pairing* gives a factor, and crossing lines add a minus sign. In our system $\langle \lambda | c_{\vec{k}, \sigma}^{\dagger} c_{\vec{k}', \sigma'} | \lambda \rangle = \delta_{\vec{k}\vec{k}'} \delta_{\sigma\sigma'} n_{\sigma}(\vec{k})$.

Wick theorem in perturbation theory.

$$E_{\lambda} = \langle \lambda | H | \lambda \rangle = \sum_{\vec{k}, \sigma} \xi_k n_{\vec{k}, \sigma} - \frac{1}{2\mathcal{V}} \sum_{\substack{\vec{k}_1, \vec{k}_2 \\ \sigma_1 = \sigma_2}} n_{\sigma_1}(\vec{k}_1) n_{\sigma_2}(\vec{k}_2) \tilde{U}(\vec{k}_1 - \vec{k}_2) + E_H, \quad E_H = \frac{N\rho \tilde{U}(0)}{2},$$

where the second term is the **exchange** (Fock) interaction and E_H is the **Hartree energy**. Despite having the expression of the energy, the ground state is not specified.

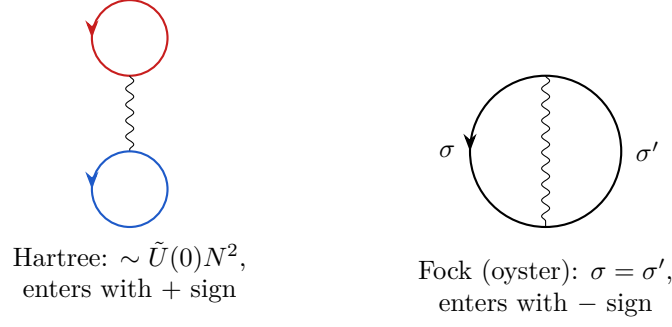


Figure 11.1: The two pairings of the interaction vertex. In the Hartree pairing there are no crossing lines; in the Fock pairing the fermion line is continuous and the sign is negative.

11.3 The ground state

For weak enough interactions the ground state remains the Fermi sea, $|\text{GS}\rangle = |\text{FS}\rangle$. Elementary excitations have energies

$$\tilde{\xi}_{\vec{k},\sigma} = \frac{\delta E}{\delta n_{\vec{k},\sigma}} = \xi_{\vec{k}} + \rho \tilde{U}(0) - \frac{1}{\mathcal{V}} \sum_{\vec{k}'} \tilde{U}(\vec{k} - \vec{k}') n_{\sigma}(\vec{k}').$$

11.4 Generalisation: the Landau theory

For strong interactions one writes, in terms of quasiparticle occupation numbers $n_{\vec{k}\sigma} \in \{0, 1\}$ (eigenvalues of $\hat{n}$, not $c^\dagger c$ of the bare particles),

$$H = \sum_{\vec{k},\sigma} \xi_{\vec{k},\sigma} n_{\vec{k},\sigma} + \sum_{\vec{k},\sigma} \sum_{\vec{k}',\sigma'} f_{\sigma\sigma'}(\vec{k}, \vec{k}') n_{\vec{k},\sigma} n_{\vec{k}',\sigma'}.$$

The angular harmonics of f (by symmetry, it depends only on the angle θ between $\vec{k}$ and $\vec{k}'$ on the Fermi surface) are the **Landau parameters**:

$$\nu_F f_{\sigma\sigma'}(\vec{k}, \vec{k}') = \sum_{l=0}^{\infty} P_l(\cos \theta) (F_l^A \hat{P}_{S=0} + F_l^S \hat{P}_{S=1}).$$

Spin structure: under a spin rotation $\psi_\sigma \rightarrow w^\sigma \psi_\sigma$, $f_{\sigma\sigma'} \rightarrow w^\sigma w^{\sigma'} f_{\sigma\sigma'}$; with the spin sum rule $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$,

$$f_{\sigma\sigma'} = f_{S=0} \mathbb{1} + f_{S=1} P_{\sigma\sigma'}^{S=1},$$

the scalar ($S = 0$) part being $\sum_\sigma f_{\sigma\sigma'}$ and the vector ($S = 1$) part $f_{\sigma\sigma'} - \frac{1}{2} \sum_\sigma f_{\sigma\sigma'}$. (In the usual notation F_l^S here, the spin channel, is written F_l^a .)

11.5 The Wilson ratio

The specific heat is given by the same expression,

$$c_V = \nu_F \int_{-\infty}^{\infty} d\varepsilon \frac{(\varepsilon - \mu)^2}{4T^2} \frac{1}{\cosh^2(\frac{\varepsilon - \mu}{2T})} = \frac{\pi^2 T}{3} \nu_F,$$

with the interacting ν_F . However, the paramagnetic susceptibility is different:

$$\chi \rightarrow \frac{\chi}{1 + F_0^S},$$

where F_0^S is negative for repulsive interactions, so $\chi_{\text{rep}} > \chi_{\text{att}}$. At $F_0^S = -1$ there is a phase transition to a ferromagnet: the **Stoner instability**. It has not been observed in Fermi gases.

12 Superconductivity

- Vanishing resistivity (Kamerlingh Onnes, Leiden, 1911).
- **Meissner–Ochsenfeld effect** (1933): the magnetic field is expelled, $B = 0$ inside, i.e. $\chi = -1$ (a perfect diamagnet), measurable without putting a device inside the material.

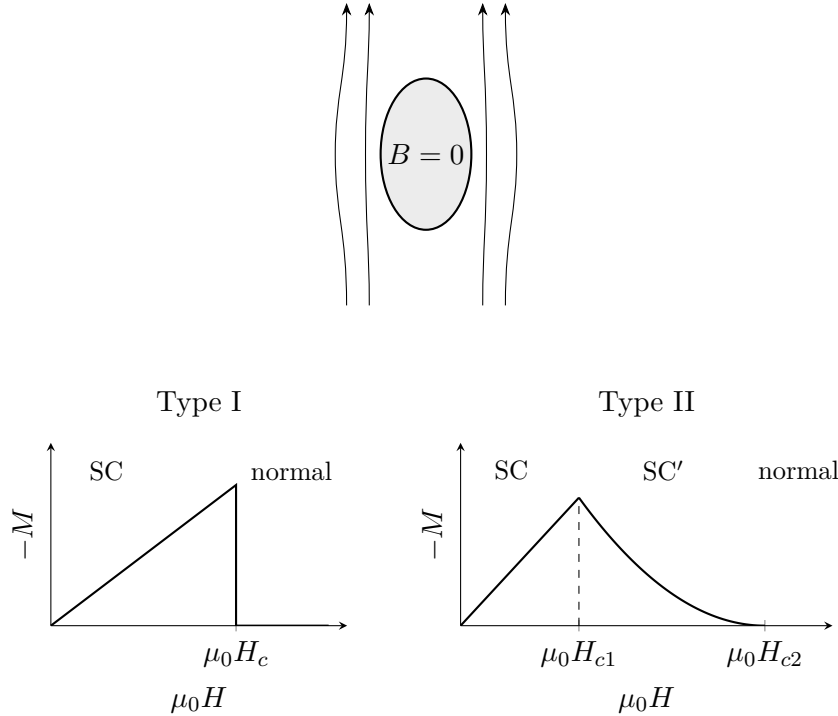


Figure 12.1: Magnetisation curves. Type II superconductors (Lev Shubnikov, 1936) have an intermediate **vortex phase** between H_{c1} and H_{c2} , in which the field penetrates as flux lines surrounded by supercurrents ($\nabla \times \vec{B} = \mu_0 \vec{j}$).

Critical magnetic field: two types of superconductors.

Heat capacity. Below T_c the heat capacity of the superconductor is suppressed at low T and jumps at T_c , compared with the linear behaviour of the normal metal.

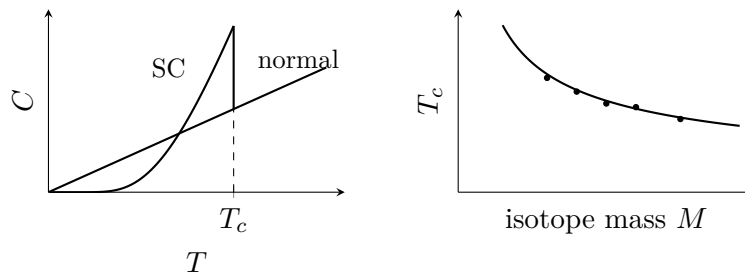


Figure 12.2: Heat capacity, and the **isotope effect** $T_c \propto M^{-1/2}$.

12.1 The Landau–Ginzburg theory

Main underlying ideas:

- vanishing resistance — similar to superfluidity;

- vortices — similar to superfluidity (“cheating”: they were not known before);
- suppression of the low- T heat capacity;
- the phase transition is a condensation? Of bosons?

GP functional for charged particles. Minimal coupling ($-i\hbar\nabla \rightarrow -i\hbar\nabla - q\vec{A}$) turns $H = \int d^3x \left(\frac{\hbar^2}{2m} |\nabla\psi|^2 + c|\psi|^4 - \mu|\psi|^2 \right)$ into

$$H = \int d^3x \left\{ \frac{\hbar^2}{2m} \left| \left(\nabla - \frac{iq}{\hbar} \vec{A} \right) \psi \right|^2 + c|\psi|^4 - \mu|\psi|^2 + \frac{B^2}{2\mu_0} \right\}, \quad \mu = b(T_c - T).$$

No Goldstone mode (Higgs mechanism). In the linear σ -model approximation, $\psi = \sqrt{\rho_s} e^{i\theta}$,

$$H = \int d^3x \left[\frac{\gamma}{2} \left(\nabla\theta - \frac{q}{\hbar} \vec{A} \right)^2 + \frac{B^2}{2\mu_0} \right], \quad \gamma = \frac{\hbar^2}{m} \rho_s.$$

The Goldstone mode is absorbed into the gauge transformation of the vector potential, $\vec{A} \rightarrow \vec{A} + \frac{\hbar}{q} \nabla\theta$.

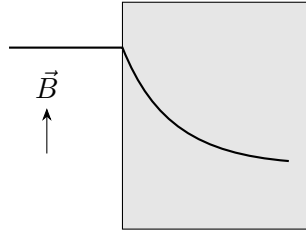
Penetration length. In the gauge where θ is absorbed,

$$H = \int d^3x \left[\frac{\gamma}{2} \left(\frac{q\vec{A}}{\hbar} \right)^2 + \frac{B^2}{2\mu_0} \right].$$

$\delta H / \delta \vec{A} = 0$ gives

$$\frac{\nabla \times (\nabla \times \vec{A})}{\mu_0} = -\gamma \left(\frac{q}{\hbar} \right)^2 \vec{A} \implies \nabla^2 \vec{B} = \gamma \mu_0 \left(\frac{q}{\hbar} \right)^2 \vec{B} = \frac{\vec{B}}{\lambda^2}, \quad \lambda = \frac{1}{\sqrt{\gamma (q/\hbar)^2 \mu_0}},$$

the **London penetration length**: the magnetic field does not penetrate deeper than λ .



Vortex. With $|\psi|^2 = \rho_s$ fixed,

$$\mathcal{H} = \frac{\gamma}{2} \left(\nabla\theta - \frac{q}{\hbar} \vec{A} \right)^2 + \frac{B^2}{2\mu_0}, \quad \theta \in \mathbb{R}/2\pi\mathbb{Z}, \quad \theta = n\phi.$$

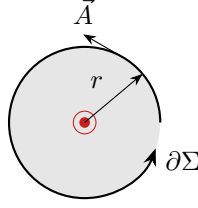
Far from the core the energy is minimised by $\nabla\theta = \frac{q}{\hbar} \vec{A}$, so $B = \nabla \times \vec{A} = 0$ there, and the flux through a surface Σ bounded by such a contour is

$$\int_{\Sigma} d^2x B_z = \oint_{\partial\Sigma} d\vec{l} \cdot \vec{A} = \frac{\hbar}{q} \oint_{\partial\Sigma} \nabla\theta \cdot d\vec{l} = \frac{\hbar}{q} 2\pi n,$$

with n the winding number:

$$\Phi_B = \frac{2\pi\hbar}{q} n = n\Phi_0,$$

with Φ_0 the **flux quantum**. Vortices with $n = \pm 1$ are favoured ($E_{n=1} < E_{n=2}$). Experimentally $q = 2e$: pairs of electrons.



13 The BCS theory

13.1 The Cooper phenomenon

Is the superconductor a condensate of electron dimers ($q = 2e$, charged bosons)? Why would electrons pair?

Cooper (1956). A particle in an attractive potential (e.g. an impurity),

$$H = -\frac{\hbar^2 \nabla^2}{2m} - g v(\vec{r}),$$

has no bound state in 3D for small g (only scattering). In the presence of the Fermi sea, in Fourier space,

$$H = \delta_{\vec{p}\vec{p}'} \frac{\hbar^2 p^2}{2m} - g \tilde{v}(\vec{p} - \vec{p}'), \quad p > p_F,$$

and there is *always* a bound state, no matter how small the coupling constant:

$$\varepsilon \sim e^{-1/(g\nu_F)},$$

which cannot be obtained by perturbation theory. The radius of the bound state is $\sim e^{1/(g\nu_F)}$. A bound state has a normalisable wave function, with energy slightly below the Fermi sea.

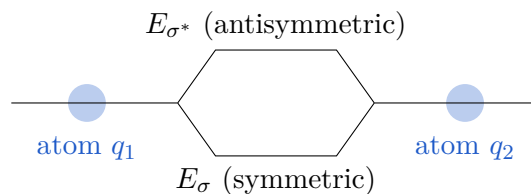
Conclusion: even a weak attraction between fermions leads to the formation of paired states with exponentially large radius.

13.2 The source of attraction

Interactions between fermions are mediated by bosons: the Coulomb interaction by photons; for electrons in a lattice, by phonons.

Toy model. Two atoms at q_1, q_2 : hybridisation of their orbitals gives a symmetric (bonding) level E_σ and an antisymmetric one E_{σ^*} (electron bands in solids are formed by hybridisation), with a splitting that depends on the distance:

$$E_{\sigma^*} - E_\sigma = \Delta(r) = \Delta_0 + \Delta'(q_1 - q_2).$$



In second quantisation (a toy model of lattice vibrations coupled to electrons),

$$H = c_{\sigma}^{\dagger} c_{\sigma} E_{\sigma} + c_{\sigma^*}^{\dagger} c_{\sigma^*} (E_{\sigma^*} + \gamma(q_1 - q_2)),$$

where $\gamma(q_1 - q_2)$ is the excitation energy as a function of $q_1 - q_2$ ($\gamma = \Delta'$). The coupling depends only on relative displacements (translation invariance), i.e. on $\nabla \vec{q}$ in the continuum — like a bound charge $\rho_b \sim \nabla \cdot \vec{P}$ arising from a polarisation.

13.3 The electron–phonon interaction

Assumptions: for simplicity, acoustic phonons only; the coupling vanishes when the wavelength of the phonon goes to infinity; the coupling is isotropic and local.

The minimal Hamiltonian. $H = H_0 + H_{e-ph}$ with

$$H_0 = \sum_{\vec{k}} \xi_k c_{\vec{k},\sigma}^{\dagger} c_{\vec{k},\sigma} + \sum_{\vec{q}} \hbar \omega_{\vec{q}} a_{\vec{q}}^{\dagger} a_{\vec{q}},$$

and the minimal interaction

$$H_{e-ph} = \gamma \int d^3x [\psi_{\sigma}^{\dagger}(x) (\vec{\nabla}) \psi_{\sigma}(x)] \cdot \nabla \theta(x), \quad \psi^{\dagger} \vec{\nabla} \psi := \frac{1}{2} (\psi^{\dagger} \nabla \psi - (\nabla \psi^{\dagger}) \psi),$$

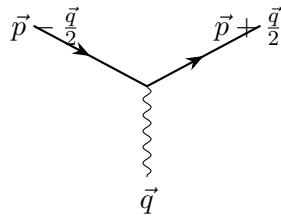
with θ the phonon displacement field,

$$\nabla \theta(x) = \sum_{\vec{q} \in \mathcal{B}} i\vec{q} \sqrt{\frac{\hbar}{2Nm\omega_{\vec{q}}}} [a_{\vec{q}} e^{i\vec{q} \cdot \vec{x}} - a_{\vec{q}}^{\dagger} e^{-i\vec{q} \cdot \vec{x}}].$$

(Integrating by parts, $\int \psi^{\dagger} (i\nabla) \psi = - \int (i\nabla \psi^{\dagger}) \psi$, so the symmetrised form is Hermitian.) In Fourier components,

$$H_{e-ph} = i \frac{\gamma'}{\sqrt{\mathcal{V}}} \sum_{\vec{p}, \vec{q}} v_{\vec{p}, \vec{q}} c_{\vec{p}+\vec{q}/2, \sigma}^{\dagger} c_{\vec{p}-\vec{q}/2, \sigma} [a_{\vec{q}} - a_{-\vec{q}}^{\dagger}], \quad v_{\vec{p}, \vec{q}} = \frac{\vec{p} \cdot \vec{q}}{\sqrt{\omega_{\vec{q}}}},$$

where constants have been absorbed into γ' .



13.4 The effective interaction strategy

For $H = H_0 + \gamma V$, the Hamiltonian can be decoupled to leading order in the perturbation parameter by a unitary transformation (Schrieffer–Wolff), $\tilde{H} = e^{-i\gamma S} H e^{i\gamma S}$ with S Hermitian:

$$\tilde{H} = H_0 + \gamma V + i\gamma [H_0, S] + i\gamma^2 [V, S] - \frac{\gamma^2}{2} [[H_0, S], S] + \dots$$

Choosing S to cancel the leading order, $[H_0, S] = iV$,

$$\tilde{H} = H_0 + \frac{i\gamma^2}{2} [V, S],$$

which is of second (or zeroth) order in the phonon displacement.

The generator of the unitary rotation.

$$S = \frac{1}{\sqrt{V}} \sum_{\vec{p}, \vec{q}} c_{\vec{p}+\vec{q}/2, \sigma}^\dagger c_{\vec{p}-\vec{q}/2, \sigma} [s_{\vec{p}, \vec{q}} a_{\vec{q}} + s_{\vec{p}, -\vec{q}}^* a_{-\vec{q}}^\dagger].$$

With $\Delta\varepsilon_{\vec{p}, \vec{q}} = \xi_{\vec{p}+\vec{q}/2} - \xi_{\vec{p}-\vec{q}/2} = \hbar v_F \hat{p} \cdot \vec{q}$ (assuming a parabolic dispersion, $\vec{p}$ on the Fermi surface),

$$s_{\vec{p}, \vec{q}} = -\frac{v_{\vec{p}, \vec{q}}}{\Delta\varepsilon_{\vec{p}, \vec{q}} - \hbar\omega_{\vec{q}}}, \quad v_{\vec{p}, -\vec{q}} = -v_{\vec{p}, \vec{q}}, \quad \Delta\varepsilon_{\vec{p}, -\vec{q}} = -\Delta\varepsilon_{\vec{p}, \vec{q}}.$$

Then

$$H_{\text{int}} = \gamma^2 \sum_{\vec{q}, \vec{p}, \vec{p}'} \frac{v_{\vec{p}\vec{q}} v_{\vec{p}'\vec{q}} \hbar\omega_{\vec{q}}}{\Delta\varepsilon_{\vec{p}\vec{q}}^2 - \hbar^2\omega_{\vec{q}}^2} c_{\vec{p}'+\vec{q}/2, \sigma'}^\dagger c_{\vec{p}-\vec{q}/2, \sigma}^\dagger c_{\vec{p}+\vec{q}/2, \sigma} c_{\vec{p}'-\vec{q}/2, \sigma'}.$$

The scales are $\Delta\varepsilon_{\vec{p}\vec{q}} \sim \hbar v_F q$ with $v_F \sim 10^6$ m/s, and $\hbar\omega_{\vec{q}} \sim \hbar v_s q$ with $v_s \sim 10^4$ m/s. For the processes that matter (energy transfer $|\Delta\varepsilon| < \hbar\omega_{\vec{q}}$, i.e. electrons close to the Fermi surface, with $\Delta\varepsilon_{\vec{p}, \vec{q}} = \Delta\varepsilon_{\vec{p}', \vec{q}}$) the denominator is negative and, with $v_{\vec{p}\vec{q}} = k_F \Delta\varepsilon_{\vec{p}\vec{q}} / (\hbar v_F \sqrt{\omega_{\vec{q}}})$, the result is a constant attraction:

$$H_{\text{int}} = -g \sum_{\vec{q}, \vec{p}, \vec{p}'} c_{\vec{p}'+\vec{q}/2, \sigma'}^\dagger c_{\vec{p}-\vec{q}/2, \sigma}^\dagger c_{\vec{p}+\vec{q}/2, \sigma} c_{\vec{p}'-\vec{q}/2, \sigma'}, \quad H_{\text{int}} = -g \int d^3x \psi_\uparrow^\dagger(x) \psi_\downarrow^\dagger(x) \psi_\downarrow(x) \psi_\uparrow(x).$$

Only opposite spins remain in the local form, since $\psi_\sigma(x) \psi_\sigma(x) = 0$. The interaction is *attractive*.

14 The BCS theory (continued)

(BCS, 1957: almost fifty years to understand the phenomenon discovered in 1911.)

Hamiltonian with phonon-mediated attraction.

$$H = \int d^3x \sum_{\sigma} \psi_\sigma^\dagger(x) \left[\frac{(-i\hbar\nabla)^2}{2m} - \mu \right] \psi_\sigma(x) - g \int d^3x \psi_\uparrow^\dagger(x) \psi_\downarrow^\dagger(x) \psi_\downarrow(x) \psi_\uparrow(x),$$

and, in Fourier components,

$$H = \sum_{\vec{k}} \xi_{\vec{k}} c_{\vec{k}, \sigma}^\dagger c_{\vec{k}, \sigma} - \frac{g}{V} \sum_{\vec{k}, \vec{p}, \vec{q}} c_{\vec{k}+\vec{q}, \uparrow}^\dagger c_{\vec{p}-\vec{q}, \downarrow}^\dagger c_{\vec{p}, \downarrow} c_{\vec{k}, \uparrow}.$$

14.1 The broken-symmetry state

Define the pair field $\Psi(x) = \psi_\downarrow(x) \psi_\uparrow(x)$, $\Psi^\dagger(x) = \psi_\uparrow^\dagger(x) \psi_\downarrow^\dagger(x)$. The broken-symmetry state $|\Delta\rangle$ has a non-zero **order parameter**

$$\Delta = \langle \Delta | \psi_\downarrow(x) \psi_\uparrow(x) | \Delta \rangle,$$

which violates the $U(1)$ symmetry $\psi_\sigma \rightarrow \psi_\sigma e^{i\theta}$ of the Hamiltonian.

The Nambu basis. Since the ground state does not respect particle-number conservation, use

$$\Phi = \begin{pmatrix} \psi_\uparrow \\ \psi_\downarrow \\ \psi_\uparrow^\dagger \\ \psi_\downarrow^\dagger \end{pmatrix}, \quad \Phi^\dagger = \begin{pmatrix} \psi_\uparrow^\dagger & \psi_\downarrow^\dagger & \psi_\uparrow & \psi_\downarrow \end{pmatrix}, \quad \{\Phi_a(x), \Phi_b^\dagger(y)\} = \delta_{ab} \delta(x-y),$$

with the transformation rule (which respects the particle-hole transformation)

$$\Phi^\dagger \begin{pmatrix} 0_{2 \times 2} & \mathbb{K}_{2 \times 2} \\ \mathbb{K}_{2 \times 2} & 0_{2 \times 2} \end{pmatrix} = \Phi^\top.$$

Then, up to a constant,

$$H = \frac{1}{2} \int d^3x \Phi^\dagger(x) Z \left(\frac{(-i\hbar\nabla)^2}{2m} - \mu \right) \Phi(x) - g' \int d^3x : \Phi^\dagger Z \Phi \Phi^\dagger Z \Phi :, \quad Z := \begin{pmatrix} \mathbb{K}_{2 \times 2} & 0 \\ 0 & -\mathbb{K}_{2 \times 2} \end{pmatrix}.$$

In this basis the anomalous averages $\langle \psi_\uparrow^\dagger \psi_\downarrow^\dagger \rangle$ appear as ordinary averages $\langle \Phi_a^\dagger \Phi_b \rangle$ (e.g. $\Phi_1^\dagger = \psi_\uparrow^\dagger$, $\Phi_4 = \psi_\downarrow^\dagger$).

14.2 BCS mean-field theory

$$\Delta = \langle \Delta | \psi_\downarrow(x) \psi_\uparrow(x) | \Delta \rangle, \quad \Delta(\vec{k}) = \langle \Delta | c_{\vec{k}\downarrow} c_{-\vec{k}\uparrow} | \Delta \rangle.$$

(The FFLO state violates translational invariance and allows $\langle c_{\vec{k}\uparrow} c_{\vec{k}'\downarrow} \rangle \neq 0$ with $\vec{k}' \neq -\vec{k}$; it has not been experimentally verified.) For the normalised excitations one has

$$H_{\text{BCS}} = \sum_{\vec{k}} \left[\xi_k c_{\vec{k},\sigma}^\dagger c_{\vec{k},\sigma} - g\Delta c_{\vec{k}\uparrow}^\dagger c_{-\vec{k}\downarrow}^\dagger - g\Delta^* c_{-\vec{k}\downarrow} c_{\vec{k}\uparrow} \right],$$

similar to the Bogoliubov Hamiltonian. It is diagonalised by

$$c_{\vec{k}\uparrow} = u_k d_{\vec{k}\uparrow} + v_k d_{-\vec{k}\downarrow}^\dagger, \quad c_{-\vec{k}\downarrow} = u_k d_{-\vec{k}\downarrow} - v_k d_{\vec{k}\uparrow}^\dagger$$

(+ h.c.). Unlike the Bogoliubov case, the transformation is a rotation, $u_k^2 + v_k^2 = 1$. Then

$$H = \sum_{\vec{p}} \varepsilon_{\vec{p}} b_{\vec{p}\sigma}^\dagger b_{\vec{p}\sigma}, \quad \varepsilon_{\vec{p}} = \pm \sqrt{\xi_p^2 + |g\Delta|^2}.$$

Particle conjugation. Already for $\Delta = 0$ one can define $b_{\vec{k}} = c_{\vec{k}}$ for $k > k_F$ and $b_{\vec{k}} = c_{-\vec{k}}^\dagger$ for $k < k_F$, so that $H = \sum_{\vec{k}} b_{\vec{k}}^\dagger b_{\vec{k}} |\xi_k|$ and the Fermi sea is the empty state, $|\text{FS}\rangle = |0\rangle$ ($c_{\vec{k}}^\dagger |\text{FS}\rangle = 0$, $b_{\vec{k}} |\text{FS}\rangle = 0$ for $k < k_F$). The gap opens at the Fermi surface: the spectrum $\sqrt{\xi^2 + |g\Delta|^2}$ has a minimum $|g\Delta| > 0$, so by the Landau criterion the system is a superfluid.

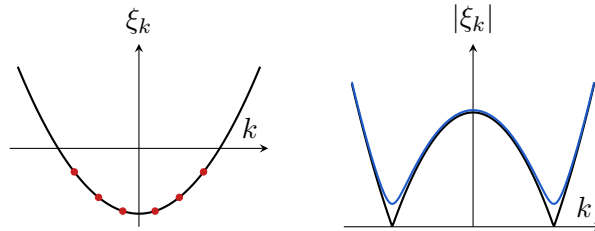


Figure 14.1: Left: occupied states of the Fermi sea. Right: after particle conjugation the excitation energies are $|\xi_k|$ (gapless at k_F); with pairing they become $\sqrt{\xi_k^2 + |g\Delta|^2}$ (blue), gapped.

14.3 Self-consistency condition

The order parameter must be reproduced by the ground state it defines:

$$\langle \Delta | c_{\vec{k}\downarrow} c_{-\vec{k}\uparrow} | \Delta \rangle = \Delta(\vec{k}) \implies H_{\text{BCS}} \implies b_{\vec{p}\sigma}, b_{\vec{p}\sigma}^\dagger \implies b_{\vec{p}\sigma} | 0 \rangle = 0 \implies \text{back to } \Delta(\vec{k}) :$$

this loop is the **BCS gap equation**,

$$\Delta = \sum_{\vec{k}} \langle \Delta | c_{\vec{k}\downarrow} c_{-\vec{k}\uparrow} | \Delta \rangle = \int_0^\Lambda dE \, \nu \frac{g\Delta}{\sqrt{E^2 + \Delta^2}},$$

with Λ the (Debye) cut-off. Without the gap, the integral would be logarithmically divergent at small E ; the divergence is truncated by Δ (at $E \sim \hbar v_F / \ell \sim \Delta$). Hence

$$1 = g\nu_F \ln \frac{\Lambda}{\Delta} \implies \Delta = \Lambda e^{-1/(g\nu_F)}.$$

A pairing of equal spins with a gradient, $\psi_\uparrow(x) \nabla \psi_\uparrow(x) = \vec{\Delta}$, gives **p-wave superconductivity**.

Revision notes

- **Sigma model:** the field is restricted to move on a manifold. From the GPE one obtains $\square\varphi = 0$, but φ is not a free field: it is an angle, $\varphi \equiv \varphi + 2\pi n$, $n \in \mathbb{Z}$ (an element of a circle rather than a line). This constrained field gives rise to vortices.
- **Motif vs basis:** mathematically the same; physically the motif is more general (a periodically repeated arrangement).
- **Validity range of the sigma model:** for $\mathcal{V}(\psi) = -\mu|\psi|^2 + g|\psi|^4$, compare the healing length ξ with the gradient of the field: trouble when $\xi^{-1} \sim \nabla\theta$, e.g. at distances $r \sim \xi$ from a vortex core, where the approximation breaks down. The sigma model (a gradient expansion) is valid if $\nabla\theta < \xi^{-1}$.
- **Debye frequency:** a cut-off chosen so that the total number of vibrational modes equals the number of atoms in the system, with $\omega_k = v_s k$.

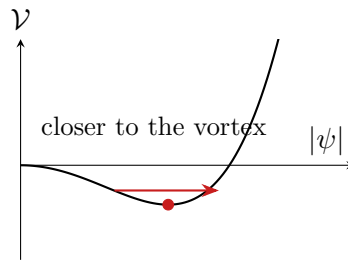


Figure 14.2: Approaching the vortex core, $|\psi|$ is pushed away from the minimum: soft (phase) fluctuations are valid only for small $\delta|\psi| \ll |\psi_0|$.